

Automating Course Articulation

A Deep Metric Learning Framework Using Public Data

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Certification of Approval

I certify that I have read Automating Course Articulation: A Deep Metric Learning Framework Using Public Data by Mark S. Kim and that in my opinion this work meets the criteria for approving a thesis submitted in partial fulfillment of the requirement for the degree Master of Science at San Francisco State University.

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Abstract

The manual process of determining course equivalency is a significant barrier to student mobility and educational equity, leading to substantial credit loss and delayed graduation that disproportionately harms underrepresented students. Previous automated approaches have been limited by a reliance on sensitive student enrollment records or the operational intractability and opacity of using large language models for direct classification.

This thesis introduces and validates a novel framework that overcomes these challenges by decoupling semantic representation from classification, using only publicly available course catalog data. The methodology leverages deep metric learning to fine-tune contextual embedding models on public course text [32]. These specialized embeddings are then used to construct a novel composite distance vector, which serves as a semantically rich feature set for training traditional machine learning classifiers such as k-Nearest Neighbors (kNN), Random Forest, and XGBoost.

Evaluated on a real-world dataset, the proposed framework achieves state-of-the-art accuracy, with F_1 -scores exceeding 0.99. The result is a computationally efficient, scalable, and privacy-preserving solution that provides institutions with a practical tool to automate course articulation, reduce administrative burden, and foster a more equitable educational ecosystem.

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Chapter 1

Introduction

California’s public higher education system is a particularly complex example of American academia which is organized as a three-tiered structure comprising of the University of California (UC), California State University (CSU), and the California Community Colleges (CCC) [17]. Collectively, these 149 colleges and universities serve nearly 2.9 million students, forming the largest public higher education system in the United States [17, 52, 11, 10]. A foundational principle of this system is the promise of student mobility, particularly the pathway from a two-year community college to a four-year university [17].

However, the mechanism designed to facilitate this movement—the process of determining course equivalency, or articulation, is a formidable, largely manual process that creates significant barriers for students [46]. At the heart of this process is the Articulation System Stimulating Interinstitutional Student Transfer (ASSIST), the state’s official public repository for articulation agreements [6]. While ASSIST provides a centralized platform for

students and advisors to view established equivalencies, it is fundamentally a repository of agreements that are negotiated and updated manually by articulation officers at each individual campus [5]. Given the sheer number of institutions, defining and maintaining these agreements is a combinatorially daunting task, rendering it inefficient, slow, and inherently intractable [46]. This manual paradigm places a considerable burden on academic advisors and administrative staff, who must meticulously review course descriptions and syllabi to compare content, rigor, and learning outcomes [46]. The result is a system that struggles to keep pace with the needs of a vast and mobile student body, making California a critical case study for a problem that extends far beyond its borders.

The challenges exemplified by California's system are a microcosm of a systemic problem in American higher education [9]. Transferring between institutions has become a normative component of the modern student's academic journey [2]. Data from the National Student Clearinghouse Research Center reveals that in the fall of 2023, transfer enrollment constituted 13.2% of all continuing and returning undergraduates [14]. This trend represents a post-pandemic resurgence in student mobility, with transfer enrollment growing by 5.3% from Fall 2022 to Fall 2023 and an additional 4.4% in Fall 2024 [14, 15]. This mobile population is increasingly diverse, comprising not only students following traditional two-year to four-year pathways but also a substantial number of returning learners [16]. Over half of these returning students opt to re-enroll at a new institution, underscoring the critical role of the transfer system in providing flexible pathways to degree completion [16].

The consequences of this systemic inefficiency are borne almost entirely by students,

manifesting in significant academic and financial setbacks [64]. The most direct outcome is the loss of earned academic credit. A 2017 report by the U.S. Government Accountability Office (GAO) estimated that students who transferred between 2004 and 2009 lost an average of 43% of their credits [64]. This finding is echoed across studies, with reports indicating that more than half of all transfer students lose some credits, and approximately one-fifth are forced to repeat courses for which they have already received a passing grade [2].

This credit loss creates a cascade of negative consequences. It invariably increases the time-to-degree, delaying graduation and entry into the workforce [64]. Each repeated course also carries a financial cost, increasing the total tuition burden and potentially exhausting a student's eligibility for federal financial aid [64]. A process often undertaken to save money can paradoxically result in a greater overall financial commitment [9]. The frustration inherent in the transfer process also has a measurable impact on student persistence, with transfer students tending to have lower graduation rates than their non-transfer peers [50].

This issue transcends mere administrative inefficiency and raises important concerns about educational equity [9]. Low-income students and students from historically under-represented racial and ethnic groups are more likely to begin at community colleges and rely on transfer pathways [63]. Recent growth in transfer enrollment has been driven disproportionately by Black and Hispanic students [14]. Therefore, the barriers imposed by an inefficient articulation system disproportionately harm the very student populations that many institutions are striving to support. A clear and troubling feedback loop emerges: the manual nature of course articulation causes credit loss, which imposes academic and

financial burdens that disproportionately affect underrepresented students, undermining institutional goals of closing equity gaps. Thus, the seemingly low-level administrative task of determining course equivalency becomes one consequential driver of systemic inequity in higher education.

Addressing this challenge through robust automation is not merely an operational optimization; it is a necessary intervention to foster a more equitable, efficient, and supportive educational ecosystem. This thesis confronts this problem by developing and validating a novel computational framework to automate the course articulation process. By leveraging deep metric learning on publicly available course catalog text, this research introduces a privacy-preserving and scalable method that decouples semantic representation from classification [32].

To achieve the above goal, this thesis project was carefully designed and include the following key tasks:

1. Evaluate the feasibility of using large language models (LLMs) for direct course equivalency classification based solely on catalog descriptions.
2. Design and validate a decoupled deep metric learning framework that separates semantic representation from classification to improve scalability and transparency.
3. Demonstrate that public course catalog data can be used to achieve high-accuracy course articulation without reliance on private enrollment or performance records.

4. Analyze classification errors to diagnose systemic limitations in the source data and identify areas for future refinement.

To fulfill these tasks, this research followed a multi-phase methodological approach. The investigation began with an exploratory feasibility study using direct classification using Google’s Gemini v1.0, to establish a performance baseline and identify the operational challenges of prompt engineering and in-context learning for this task. The insights from this initial phase motivated the development of the primary contribution: a decoupled pipeline framework designed for scalability and transparency.

The core of this framework involves several key implementation steps. First, it generates semantic representations of course data using a suite of pre-trained sentence embedding models, including BGE, GIST, and NVE (see Section 3.3). To adapt these general-purpose models to the specific domain of course description text, a supervised fine-tuning process is employed using a deep metric learning objective [32]. This creates specialized embeddings that are then used in a novel feature engineering step to construct a composite distance vector for each course pair. This process is followed by an evaluation of dimensionality reduction techniques to optimize the feature set. Finally, these features are used to train and evaluate a range of traditional, lightweight machine learning classifiers, such as Support Vector Machines and Random Forest, to perform the final equivalency prediction. The entire pipeline is subjected to a rigorous evaluation and error analysis to validate its effectiveness.

1.1 Thesis Roadmap

The remainder of this thesis is structured to provide a comprehensive account of this research.

- **Chapter 2: Background and Related Work** will provide a detailed in-depth survey of the landscape of student transfer automation and the evolution of technological interventions.
- **Chapter 3: Methodology** will offer a deep dive into the data collection and preparation processes, the specific embedding models evaluated, the construction of the feature vectors, and the theoretical underpinnings of the machine learning classifiers employed.
- **Chapter 4: Experimental Setup and Results** will detail the experimental design, the datasets used for training and validation, and a comprehensive analysis of the classification performance, including ablation studies and model comparisons.
- **Chapter 5: Discussion, Future Work, and Conclusion** will interpret the results in a broader context, discuss the limitations of the current study, outline promising avenues for future research, and conclude with a summary of the key findings of the thesis and reiterate the significance of its contributions to both academic research and the practical administration of higher education.

1.2 Contributions and Significance

This thesis contributes a novel, empirically validated approach to the automation of course articulation in higher education. Its core contributions are:

1. A scalable, privacy-preserving articulation framework that relies solely on publicly available course catalog data—eliminating the dependence on sensitive enrollment records or opaque LLM inference pipelines.
2. A composite distance feature vector that integrates both local (elementwise) and global (cosine similarity) semantic measures between course embeddings, improving classification accuracy over standard approaches.
3. A fine-tuning protocol for deep metric learning applied to course text embeddings, shown to outperform even large-scale off-the-shelf models on real-world equivalency classification tasks.
4. A rigorous, multi-phase evaluation including ablation studies, statistical significance testing, and qualitative misclassification analysis that collectively demonstrate the proposed framework’s effectiveness and generalizability.
5. A practical pathway towards institutional deployment, including insights on model efficiency, error patterns, and a human-in-the-loop extension for building trust and improving training data over time.

Together, these contributions provide not only a methodological innovation but also a practical tool to advance equity, reduce administrative burden, and scale articulation processes across diverse academic systems.

Chapter 2

Background and Related Work

The pronounced challenges of manual course articulation, detailed in the previous chapter, have prompted a range of research efforts aimed at automating the process [38, 47, 46, 30, 73]. These technological interventions have evolved in sophistication, mirroring broader advancements in natural language processing (NLP) and machine learning [60]. A critical review of this literature reveals a clear trajectory from simple statistical methods to complex deep learning models, with each stage introducing new capabilities while also exposing new limitations [46]. This evolution illuminates the path toward a more robust and scalable solution.

2.1 Keyword and Statistical Methods

The earliest attempts at automating course comparison relied on foundational text analysis techniques that, while computationally simple, lack semantic depth. The most basic systems are essentially search engines or databases that depend on exact keyword matching or pre-populated tables of known equivalencies [35]. These systems are inherently brittle; they cannot recognize semantic variations (e.g., equating “Introduction to Programming” with “Fundamentals of Computer Science I”) and require continuous manual updates to remain relevant [60].

A more advanced statistical method, Term Frequency-Inverse Document Frequency (TF-IDF), improves upon keyword matching by vectorizing documents and weighting terms based on their importance. A term’s frequency within a single document (TF) is balanced against its rarity across a corpus (IDF) [3]. This allows the model to assign higher importance to distinctive terms (e.g., “calculus”) and lower importance to common words (e.g., “the,” “a,” “is”) [3]. While TF-IDF has been a workhorse for information retrieval, its fundamental limitation is a complete lack of semantic understanding [3]. Models in this class treat words as discrete, unrelated tokens; they cannot grasp that “calculus” and “differentiation” are related concepts, nor can they distinguish between different meanings of the same word.

2.2 Static Semantic Representations

The development of word embeddings represented the first major leap toward a true semantic understanding of course content. Models like Word2Vec and GloVe, trained on vast text corpora, learn to represent words as dense vectors where words with similar meanings are positioned closer to one another in the vector space [41, 49]. This innovation enabled a more nuanced comparison of texts than was possible with TF-IDF. These techniques have been applied to content-based course recommendation by creating a single vector for a course description, typically by averaging the vectors of its constituent words [46].

Despite this advancement, these models produce static embeddings where each word is assigned a single, fixed vector regardless of its context [19]. This is a significant drawback, as it fails to account for polysemy—words with multiple meanings. For instance, the word “bank” would have the same vector in “river bank” and “bank account.” Furthermore, the practice of averaging all word vectors to create a document-level representation is a crude heuristic that can dilute or lose crucial semantic information [53].

2.3 Contextual Semantic Representations

The introduction of the transformer architecture, specifically models like Bidirectional Encoder Representations from Transformers (BERT), revolutionized NLP by enabling the generation of contextual embeddings [19]. In these models, a word’s vector representation is dynamically influenced by the words surrounding it, allowing the model to disambiguate

word meanings and capture a much richer semantic representation [19]. Architectures such as Sentence-BERT (SBERT) were subsequently developed to fine-tune these models to produce semantically meaningful embeddings for entire sentences or paragraphs, which can then be efficiently compared using metrics like cosine similarity [53].

2.4 Direct LLM Classification

A more recent evolution involves the direct application of large-scale generative models, or Large Language Models (LLMs) like GPT-4 and Gemini, for classification. As explored in preliminary work for this thesis, these models can be instructed via prompt engineering and in-context learning to perform pairwise comparisons of course descriptions and render a judgment on their equivalency. While these experiments yielded promising accuracy, they also uncovered considerable practical limitations.

The direct use of proprietary, API-based LLMs for this task is operationally and economically challenging. Each pairwise comparison requires a separate API call, which incurs both financial cost and notable latency [59, 44]. For a system that needs to compare thousands of courses across hundreds of institutions, the cumulative cost and time would be prohibitive [59, 44]. Furthermore, to ensure a responsive user experience in an interactive setting, end-to-end latency must be kept low, a target that is difficult to meet with large models under heavy load [44, 59]. This makes the direct classification approach impractical for a scalable, production-level system and justifies the pivot to a more efficient, decoupled

framework that performs the expensive semantic processing once (embedding generation) and then uses lightweight classifiers for rapid comparison.

Likewise, performance is acutely sensitive to prompt phrasing, a finding that aligns with a large and growing body of academic research that identifies this as a systemic challenge for LLMs [51, 81]. Studies have formally demonstrated that even minor, semantically-equivalent changes in prompt wording can lead to drastically different outputs [51, 81]. This brittleness and lack of robustness pose a formidable challenge for deploying LLMs in high-stakes, operational systems like course articulation, where consistency and reliability are paramount. A system that produces different equivalency judgments based on trivial variations in prompt formatting is not a trustworthy tool for students or administrators, which also makes it ill-suited for handling complex one-to-many or many-to-many articulation scenarios. This opacity makes it difficult to rank matches, set thresholds, or provide transparent justifications.

2.5 Enrollment-Based Approaches

Parallel research efforts have leveraged different data sources entirely. A notable body of work has demonstrated that course similarity can be predicted by analyzing student enrollment data [47, 30]. Models such as *course2vec* learn embeddings from the patterns of which courses students take together, operating on the principle that courses taken in the same semester or sequence likely share a topical relationship [47].

While powerful, this behavioral approach is constrained by its reliance on large-scale, proprietary institutional datasets. This raises considerable data privacy concerns, as the use of such data is fraught with ethical and legal challenges, including compliance with regulations like the Family Educational Rights and Privacy Act (FERPA) in the United States [54]. The collection and analysis of student data, even for beneficial purposes, raises concerns about unauthorized profiling, potential for discrimination, and the risk of data breaches [54].

Beyond data privacy, however, a more fundamental technical limitation exists that is vital to the specific problem of inter-institutional articulation: the cold-start problem [7, 78, 1]. Enrollment-based methods are a form of collaborative filtering, a technique common in recommender systems [7]. A well-documented weakness of collaborative filtering is its inability to handle new users or new items for which it has insufficient interaction data [7, 78, 1]. This problem manifests directly and severely in the course articulation context. When a community college creates a new course, or when a university has no history of student transfers from a particular feeder institution, there are no co-enrollment patterns to learn from. The model is effectively blind. This represents a fundamental limitation for the exact use case of establishing new articulation agreements. Therefore, these models suffer from a pivotal generalizability crisis when applied to the inter-institutional articulation discovery task, which potentially requires comparing institutions with sparse or non-existent transfer histories. This technical gap underscores the need for a framework that relies on public data, which is universally available.

2.6 The Search for an Alternative

This review of varied approaches reveals a primary trade-off: as models gain semantic power, they tend to become more computationally intensive, less interpretable, or more demanding of specialized or private data. The limitations of direct LLM classification (cost, opacity) and enrollment-based methods (data privacy, limited access) point toward a gap in the existing research for a new paradigm [46]. An effective solution must harness the semantic power of large pre-trained models without inheriting their operational burdens. The literature thus indicates a need for an effective, hybrid framework that decouples deep semantic representation from final classification. This conceptual gap forms the central motivation for the methodology proposed in this thesis. Table 2.1 summarizes the primary methods for determining course transferability.

Table 2.1: Comparative Taxonomy of Course Equivalency Determination Methods

Approach	Key Characteristics	Data Source(s)	Semantic Capability	Strengths	Limitations
Manual Review	Human experts (advisors, faculty) compare syllabi descriptions.	Course Catalogs, Syllabi	High (Human-level)	Nuanced, context-aware, trusted by faculty.	Extremely slow, not scalable, subjective, prone to inconsistency.
Keyword/TF-IDF [30]	Bag-of-words representation, statistical term weighting.	Course Catalogs	None to Low	Simple, computationally cheap, easy to implement.	Fails to capture synonyms, context, or true semantic meaning
Static Embeddings (Word2Vec/GloVe) [46]	Pre-trained word vectors, often averaged for document representation.	Course Catalogs	Medium	Captures word-level semantics, better than TF-IDF.	Context-insensitive, averaging vectors loses information.
Enrollment-Based (e.g., course2vec) [47]	Embeddings learned from student co-enrollment patterns.	Proprietary Student Records	High (Behavioral)	Captures functional relationships between courses, highly predictive.	Requires access to sensitive private data, not generalizable, privacy concerns.
Direct LLM Classification	End-to-end classification using prompt engineering.	Course Catalogs	Very High	High accuracy potential, understands complex language.	Computationally expensive, “black box” opacity, prompt sensitive, no quantifiable similarity score, risk of hallucinations.
Proposed Method (Embeddings + ML)	Deep contextual embeddings as features for traditional classifiers.	Course Catalogs	Very High	State-of-the-art accuracy, computationally efficient, quantifiable, uses public data only.	Relies on the quality of the pre-trained embedding model.

Chapter 3

Methodology

This chapter delineates the comprehensive and systematic methodology employed to develop and evaluate a sophisticated classification pipeline capable of determining course equivalency from domain-specific textual data. As established in the preceding chapters, the manual nature of course articulation creates considerable barriers for students. Furthermore, as discussed in Chapter 2, previous automated approaches have faced significant limitations related to data privacy, scalability, and interpretability, indicating a need for a new paradigm [46, 61].

To address these challenges, the research followed an evolutionary, two-phase process. The investigation commenced with an initial, exploratory phase to assess the feasibility of using Large Language Models (LLMs) as end-to-end classifiers. The findings from this stage were informative; they demonstrated the potential of modern LLMs but also highlighted the limitations that motivated the development of a more robust and scalable framework.

Consequently, the primary focus of this thesis is a more advanced, decoupled methodology that utilizes deep embedding models as sophisticated feature extraction engines.

This chapter details this entire methodological journey. It begins by describing the initial direct LLM classification approach and the findings that motivated the subsequent pivot. It then details the final pipeline’s core components: the data corpora, the feature engineering process, the selection and fine-tuning of deep embedding models, and the suite of downstream classifiers. Finally, by outlining the complete evaluation framework, including performance metrics, statistical analyses, and error analysis methodologies, this chapter sets a clear and rigorous stage for the discussion of results in Chapter 4.

3.1 Phase 1: Direct LLM Classification Approach

The research commenced with an exploratory phase designed to determine the feasibility of leveraging Large Language Models (LLMs) as end-to-end classifiers for the course equivalency task. This direct approach was a pragmatic first step, conceived as an initial benchmark to establish a baseline for performance using a small, manually curated dataset. At this stage of the research, a larger, more comprehensive dataset from the Program Pathways Mapper ¹ (PPM) program was anticipated but not yet available, making a focused, smaller-scale investigation a reasonable starting point given the available data. This methodology treated

¹Program Pathways Mapper (PPM) is a tool for academic planning and promoting program completion. Having received California State funding, PPM has been able to acquire data drawn from UC, CSU, CCC, and the Chancellor’s Office Curriculum Inventory (COCI) to create a “golden record or canonical data set.” For more information, refer to <https://programmapper.org/>

the LLM as a holistic reasoning engine, tasked with performing the entire classification from raw text input to a final equivalency judgment without intermediate feature engineering.

Initial Data Corpus and Pre-processing

The dataset for this initial evaluation was constructed to represent a challenging, real-world scenario using publicly available data. The process began by identifying five required lower-division courses for the Computer Science major at San Francisco State University (SFSU). Using ASSIST, articulation agreements were found for these courses across 63 different California public colleges and universities. The raw course data was then manually collected from the online course catalogs of each respective college. This data consisted of the full, unmodified text including department codes, course numbers, titles, descriptions, and all associated metadata such as prerequisites, unit counts, and grading options (see Table 3.1). This approach was deliberately chosen to ensure that the analysis could compare the effectiveness of classification using the complete raw text versus more structured, extracted information.

Table 3.1: Raw Course Text Examples

MATH 226 Calculus I (Units: 4) Prerequisite: One of the following: MATH 198 or MATH 199 or equivalent with a grade of C or better; or MATH 226 or equivalent with a grade of C- or lower; or high school pre-calculus with B or better; or high school calculus with a grade of C or better. Graphs. Differentiation: theory, techniques, and applications. Integration: Fundamental Theorem of Calculus and applications. Transcendental functions. Lecture, 3 units; laboratory, 1 unit. Course Attributes: 2: MATH/Quantitative Reason B4: Math/QR
CSC 256 Machine Structures (Units: 3) Prerequisite: CSC 230 with a grade of C or better. Digital logic circuits, data representation, assembly language programming, subroutine linkage, machine language encoding, interrupt and exception handling, memory system concepts, and CPU organization and performance.

The initial dataset consisted of 228 equivalent course pairs based on the articulation agreements (see Table 3.2). To create a more robust dataset for binary classification, this set was expanded by assuming symmetry and transitivity for course equivalency, which generated a total of 5,660 equivalent pairs. An equivalent number of non-equivalent pairs was then generated by randomly pairing courses from different subjects. From this expanded corpus of over 11,000 pairs, a final stratified random sample of 400 pairs (200 equivalent and 200 non-equivalent) was created to serve as the evaluation set for the models.

Model Selection and Prompt Engineering

A preliminary review of various LLMs was conducted to assess their ability to reliably generate structured data from the raw course descriptions. While many open-source and proprietary models were tested, Google’s PaLM2 and its successor, Gemini Pro v1.0, were

Table 3.2: SFSU Courses Identified

Code^a	Title	Count
MATH 226	Calculus I	61
CSC 210	Intro to Computer Programming	51
CSC 220	Data Structures	44
CSC 230	Discrete Math	47
CSC 256	Machine Structures	25

^aCourse code from San Francisco State University course catalog

ultimately selected for this phase of the research [45, 42]. This decision was based on their accessibility via a free-tier API and, most importantly, their consistent ability to produce well-formatted, structured data from the unprocessed text.

A structured, iterative prompt engineering process, illustrated in Figure 3.1, was employed to develop effective prompts for both the extraction of structured data (like course topics) and the final equivalency classification. This process involved starting with simple prompts and gradually refining them based on established design principles from natural language processing research and community guides [75, 34, 55]. The final prompts for data extraction were highly structured, consisting of five parts: a preamble summarizing the task, the raw course data, specific formatting instructions, a JSON model schema defining the desired output, and a postamble with additional clarifying instructions. A complete prompt template examples can be found in Appendix A. Despite this careful refinement, the structured data extraction process, particularly for deducing course topics, remained a notable challenge and was prone to occasional contextual errors.

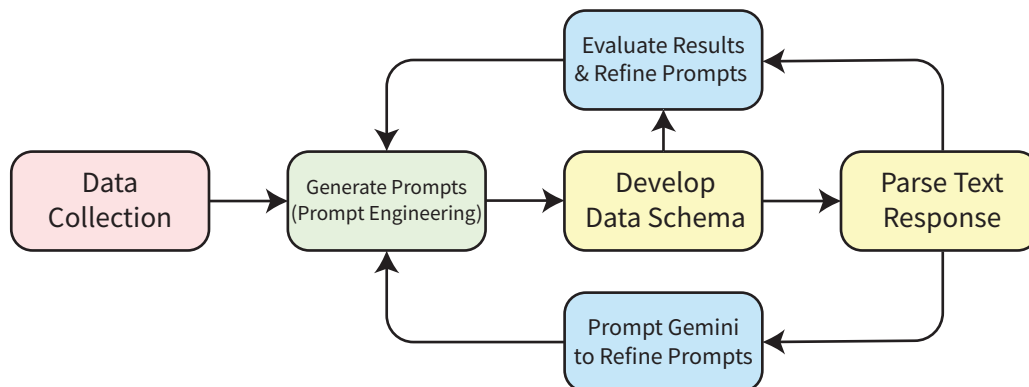


Figure 3.1: Prompt Engineering Process

3.2 Phase 2: The Decoupled Pipeline Framework

The limitations identified in the initial study—including high computational cost, lack of quantifiable similarity scores, and prompt sensitivity—directly motivated a pivot to a more sophisticated, decoupled pipeline. This new approach was first developed and prototyped using the initial, manually-curated dataset. This section details this final, more robust methodology, beginning with the larger and more comprehensive data corpus upon which it was ultimately trained and validated. The complete pipeline workflow is illustrated in Figure 3.2.

The PPM Corpus and Data Preparation

The limitations of the initial study necessitated not only a more sophisticated methodology but also a larger, more comprehensive dataset to robustly train and evaluate it. This required data corpus was procured at a later stage of the research in partnership with the Program Pathways Mapper (PPM). The acquisition of this dataset was a critical step that

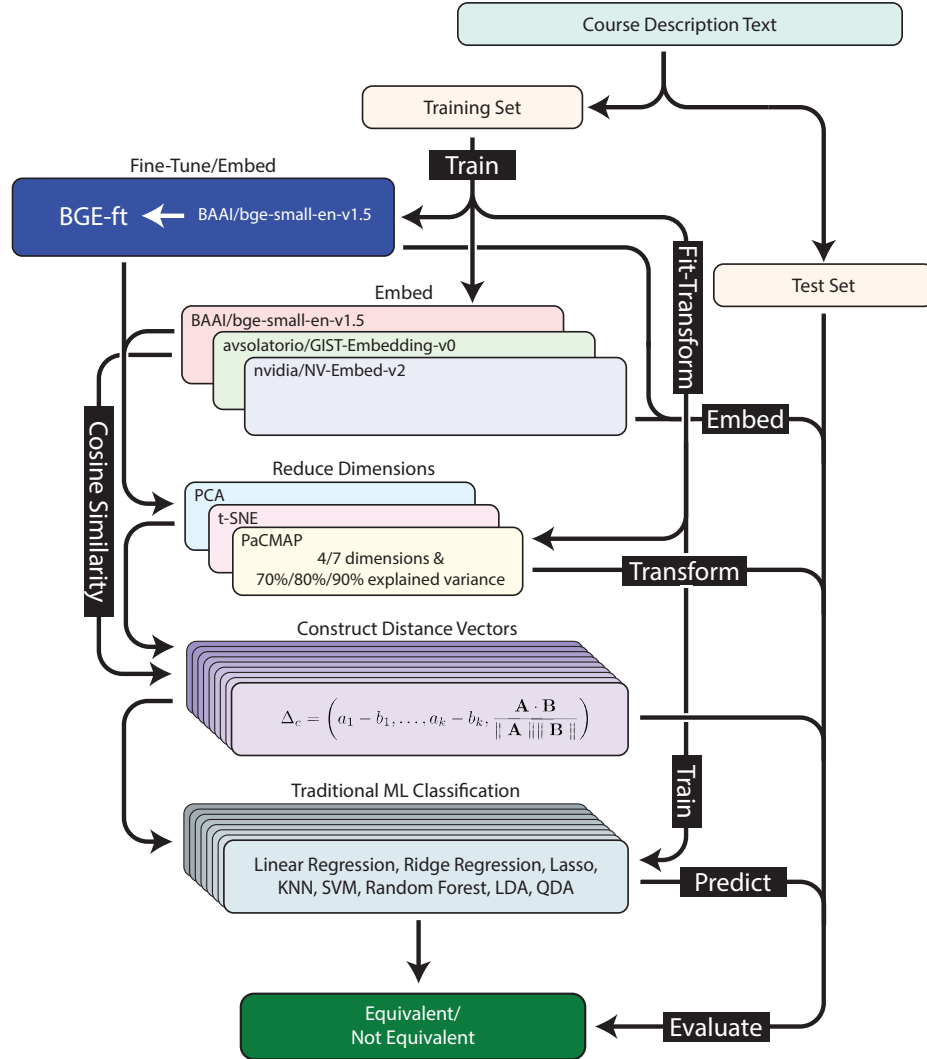


Figure 3.2: The Decoupled Pipeline Framework

enabled the full-scale implementation and validation of the decoupled pipeline.

The corpus provided by the PPM initially contained 2,217 courses, each labeled with a Course Identification Numbering System ² (C-ID) code that serves as the ground truth for course equivalency. To prepare this data for a robust, stratified partitioning, a critical

²The Course Identification Numbering System (C-ID) is a common course numbering system being developed by the Academic Senate for California Community Colleges to provide a mechanism to identify comparable courses among colleges. For more information, refer to <https://c-idssystem.org/>

Table 3.3: PPM Data Split

Data Set	Number of Course Pairs		Total
	Equivalent	Non-Equivalent	
Training	4336	4336	8672
Test	4596	4596	9192

filtering step was applied first. All C-ID classes with fewer than four associated courses were removed from the dataset. This step was essential to guarantee that after splitting the data, both the training and subsequent test sets would contain enough examples to form at least one equivalent course pair for every class, a necessary condition for the fine-tuning process. This filtering resulted in a final, clean corpus of 2,157 courses distributed across 157 distinct C-ID classes.

To ensure a rigorous and unbiased evaluation of the final models, this final dataset was partitioned into two distinct, non-overlapping subsets: a training set and a test set. A stratified 50/50 split was employed, using the C-ID code as the stratification key. This resulted in a training set of 1,078 courses and a test set of 1,079 courses. The stratification ensures that each of the 157 classes is represented in both subsets. From the training and test sets, a total of 8672 and 9192 course pairs were generated, which had a 50/50 split of equivalent and non-equivalent course pairs (see Table 3.3). The test set is held in reserve and used only once for the final, conclusive evaluation of the optimized classification pipeline, providing an honest estimate of the model’s generalization performance on unseen data.

Input Document Normalization

In a departure from conventional NLP pipelines, this research deliberately eschewed standard text pre-processing techniques such as lowercasing, stop-word removal, or the stripping of special characters. This decision was made to more accurately simulate a real-world use case where input data may be imperfectly formatted. The methodology, therefore, relies on the inherent semantic power and robustness of modern transformer-based embedding models to interpret and handle this “raw” text.

Instead of cleaning the text, a normalization step was performed to create a consistent, structured input document for the embedding models. A new field, “Formatted Course Info,” illustrated in Table 3.4, was generated for each course by concatenating four key pieces of information: the department name, the department course number, the course title, and the full course description. This concatenated string serves as the single document representation for each course and is the direct input for the document embedding process, ensuring all relevant textual context is preserved in a standardized format.

Document Embedding

Once each course is represented by a standardized text document, the next step is to convert this textual information into a semantically rich, numerical format that machine learning models can process. This is achieved through document embedding, a process that moves beyond simple word statistics to capture the underlying semantic meaning of the text.

Table 3.4: Example of Formatted Course Info

Template	Example
$\langle$ Dept Code $\rangle$ - $\langle$ Course Code $\rangle$ $\langle$ Course Title $\rangle$ $\langle$ Course Description $\rangle$	CSC-215 Intermediate Computer Programming Prerequisite: CSC 101 with a grade of C or better. Design, implementation, testing, debugging, maintenance, and documentation of Java pro- grams. Algorithms, programming concepts, and data types in Java. Concepts of object-oriented programming. Numerical and non-numerical problems. Hands-on exercises in programming, and the use of basic software development tools.

This approach marks a significant advancement over the static word embedding methods discussed in Chapter 2, such as Word2Vec or GloVe [41, 49]. Those earlier models produce a single, fixed vector for each word, regardless of its context. This limitation makes them unable to distinguish between different meanings of the same word (polysemy) and often relies on crude heuristics like averaging word vectors, which can dilute or lose critical semantic information [30].

In contrast, this research employs pre-trained, transformer-based deep learning models. The introduction of these architectures, as detailed in Section 2.3, was transformative for Natural Language Processing because they generate contextual embeddings, which capture nuance and disambiguate meaning far more effectively. The key property of these embeddings is that their position in the vector space represents the semantic meaning of the course description; courses that are semantically similar are located closer to one another.

This research evaluates a spectrum of these modern architectures to ensure a comprehensive analysis. The models selected for in-depth evaluation include BAAI/bge-small-en-v1.5

(BGE), `avsolatorio/GIST-Embedding-v0` (GIST), and `nvidia/NV-Embed-v2` (NVE), which were chosen to represent a range of parameter sizes and designs. The specific characteristics of these models and the methodology for their selection are detailed in Section 3.3. These generated embeddings form the foundational semantic representations that are then processed in the subsequent feature engineering stage.

Feature Engineering

The high-dimensional embedding vectors generated by the models, while semantically rich, are not the final features used for classification. To prepare the data for the downstream classifiers, a two-stage feature engineering pipeline was executed. This pipeline first applies various dimensionality reduction techniques and then constructs pairwise difference vectors from these embeddings to explicitly represent the relationship between two courses.

Dimensionality Reduction

The high-dimensional vectors produced by modern embedding models can present challenges for downstream machine learning algorithms, a phenomenon often referred to as the “curse of dimensionality.” To address these potential issues, a methodical process of dimensionality reduction was applied. This investigation was motivated by several potential benefits, including improving model generalization by removing redundant or noisy dimensions and reducing computational complexity.

To ensure the integrity of the evaluation and prevent any form of data leakage, the reduction models were governed by a strict protocol. Each reduction technique was fit exclusively on the training data. The same fitted model was then used to transform both the training and the held-out test sets. This methodology guarantees that no information from the test set influences the parameters of the reduction models. A comprehensive exploration was conducted to determine the optimal dimensionality, including using Principal Component Analysis (PCA) to reduce vectors to the number of components required to explain 70%, 80%, and 90% of the original variance, as well as reducing to static 4 and 7 dimensions using PCA, t-SNE, UMAP, and PaCMAP [48, 39, 40, 70].

Composite Distance Vector

The ultimate goal of this research is to classify pairs of courses, which requires input features that represent the relationship between them. To provide a richer, more discriminative representation than a single metric alone, we designed a composite feature vector, Δ_c . The vector is constructed by concatenating the element-wise difference of the two course embedding vectors ($\mathbf{A}$ and $\mathbf{B}$) with their cosine similarity:

$$\Delta_c = \left(a_1 - b_1, \dots, a_k - b_k, \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \|\mathbf{B}\|} \right)$$

where $\mathbf{A} = (a_1, \dots, a_k)$ and $\mathbf{B} = (b_1, \dots, b_k)$ are the k -dimensional embedding vectors for the two courses. This design is powerful because it provides the subsequent classifier with two

distinct types of information simultaneously. The element-wise difference captures granular, dimension-specific (local) disparities between the two semantic representations, while the cosine similarity provides a single, normalized measure of their overall (global) alignment in the vector space. A custom `CoursePairGenerator` class, found in the project’s codebase, was implemented to generate these feature vectors for all positive and negative pairs across all embedding and reduction variations.

Overview of the Classification Stage

With the composite distance vectors constructed, the final step in the decoupled pipeline is to use these features to train a traditional machine learning classifier for the binary prediction task. Classical machine learning offers a suite of robust, computationally efficient, and highly interpretable models that are well-suited for this final classification stage.

To ensure a comprehensive evaluation, this research considered a wide array of algorithmic families. The initial investigation included linear models (e.g., Logistic Regression), instance-based models (k-Nearest Neighbors), a kernel-based model (Support Vector Machine), and powerful ensemble methods (e.g. Random Forest). This broad approach allows for a thorough exploration of the feature space to identify the most effective classification strategy. A detailed overview of these algorithmic families is presented in Section 3.3.

3.3 Model Architecture and Training

A central hypothesis of this research is that a generic, pre-trained deep embedding model can be adapted to produce highly specialized and semantically rich embeddings for the specific domain of course descriptions. These bespoke embeddings are expected to provide a more discriminative feature representation for downstream classification tasks compared to off-the-shelf pre-trained embedding models. This section details the architecture, learning objective, and training protocol used to achieve this adaptation through a process of deep metric learning, as well as the downstream classifiers used to evaluate the resulting features.

Embedding Models: Selection and Fine-Tuning

The selection of an appropriate embedding model is a critical first step that influences the entire downstream pipeline. Rather than committing to a single model, this research began with a broad preliminary analysis of a variety of open-source embedding models to identify strong candidates for a more in-depth, comparative study.

Model Selection

The initial models reviewed, summarized in Table 3.5, spanned a wide range of parameter sizes and characteristics. To screen these models efficiently, the simple but effective cosine similarity accuracy metric was used: for a given anchor course, a model was considered correct if the cosine similarity to an equivalent course was greater than the similarity

Table 3.5: Initial Embedding Model Review

Model Name	Rank*	Params [†]	Dims	Acc
GIST-small-Embedding-v0	41	33	384	0.9759
bge-small-en-v1.5	47	33	384	0.9670
GIST-Embedding-v0	33	109	768	0.9768
bge-base-en-v1.5	35	109	768	0.9732
gte-base-en-v1.5	31	137	768	0.9732
mxbai-embed-large-v1	24	335	1024	0.9759
gte-large-en-v1.5	21	434	1024	0.9777
multilingual-e5-large-inst	34	560	514	0.9670
stella_en_1.5B_v5	3	1543	8192	0.9857
SFR-Embedding-2_R	4	7111	4096	0.9839
Agte-Qwen2-7B-instruct	5	7613	3584	0.9804
nvidia/NV-Embed-v2	1	7851	4096	0.9831

* Huggingface Overall Leaderboard Rank

[†] in Millions

to a non-equivalent course. This initial screening revealed that while performance varied, many models achieved high accuracy. Based on these results and a desire to evaluate a representative spectrum of model sizes, three models were selected for the primary analysis:

- BAAI/bge-small-en-v1.5 (BGE): Representing a high-performing small model.
- avsolatorio/GIST-Embedding-v0 (GIST): Representing a medium-sized model.
- nvidia/NV-Embed-v2 (NVE): Representing a large-scale model.

At a later stage of the research, Salesforce/SFR-Embedding-2_R (SFR) was also included for additional comparison due to its strong performance on public leaderboards. The foundational architecture for all these models is the transformer, which allows them to generate rich, contextual embeddings suitable for feature extraction [65].

Metric Learning with Batch Triplet Loss Functions

To determine if performance could be further improved, the BGE model was subjected to a fine-tuning process using a deep metric learning approach. Metric learning aims to learn a direct mapping from a high-dimensional input to a compact Euclidean embedding space where the geometric distance between vectors directly corresponds to a measure of semantic similarity. This technique was famously and effectively employed by the FaceNet system [56], which used a Triplet Loss function to learn a unified embedding for face recognition. This thesis adapts this proven approach, translating it from the domain of computer vision to the domain of semantic text similarity.

The concept of Triplet Loss, first introduced in the context of face recognition, has been widely applied to supervised similarity learning [77]. It operates on (Anchor, Positive, Negative) triplets. The fundamental goal is to train the embedding function, $f(x)$, to map inputs into a vector space where the distance between an anchor sample (A) and a positive sample (P) is smaller than the distance to a negative sample (N) from a different class, enforced by a margin, α . The mathematical formulation is given by:

$$L(A, P, N) = \max(d(f(A), f(P)) - d(f(A), f(N)) + \alpha, 0).$$

Here, d represents the Euclidean distance. The $\max()$ component ensures that loss is only incurred for triplets that violate the margin constraint. A naive “offline” implementation of Triplet Loss that forms all possible triplets is computationally infeasible. A more effective ap-

proach is “online” mining, where informative triplets are selected on-the-fly from within each mini-batch. This empirical evaluation is motivated by research demonstrating that selecting the “hardest” positive and negative examples within each mini-batch is a highly effective strategy for training deep metric learning models [26]. Recognizing that different triplet mining strategies can substantially impact model performance, we empirically evaluated all four primary batch-based triplet loss implementations available in the `sentence-transformers`³ library [53]:

- **BatchAllTripletLoss**: Computes the loss for all valid triplets within a batch. While comprehensive, the learning signal can be diluted by the high number of “easy” triplets.
- **BatchSemiHardTripletLoss**: Considers only semi-hard triplets, where the negative sample is more distant than the positive but still violates the margin. This is a common strategy that balances stability and learning effectiveness.
- **BatchHardTripletLoss**: Implements a more aggressive strategy, using the hardest positive and hardest negative for each anchor. This can accelerate convergence but can also be “temperamental” and lead to a noisy optimization landscape.
- **BatchHardSoftMarginTripletLoss**: A variation of `BatchHardTripletLoss` that does not require a manually specified margin parameter, simplifying hyperparameter tuning.

³The SentenceTransformers library can be found at: <https://sbert.net/>

Optimization and Training Protocol

The successful fine-tuning of a model with a challenging objective like `BatchHardTripletLoss` is critically dependent on the choice of optimizer and learning rate schedule. The components selected for this research were not chosen in isolation but as parts of a cohesive, synergistic framework designed to ensure stable and effective learning. Figure 3.3 illustrates the interconnections of all components used during the fine-tuning lifecycle while a simplified pseudocode representation can be found in Appendix D.1.

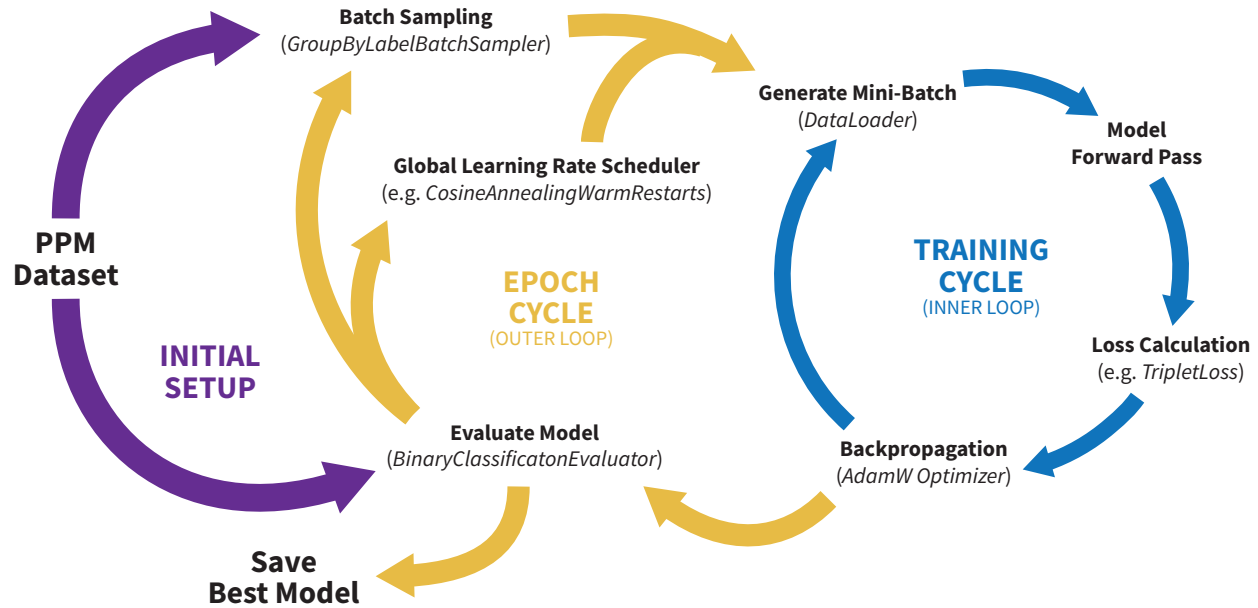


Figure 3.3: Fine-tuning Process Lifecycle

The optimization of the model’s weights was performed using the AdamW optimizer. Proposed by Loshchilov and Hutter, AdamW improves upon the standard Adam optimizer by addressing a subtle flaw in how it handles L2 regularization [36]. In the standard Adam

optimizer, weight decay is coupled with the adaptive gradient calculation, which can lead to suboptimal regularization and poorer generalization. AdamW decouples the weight decay from the gradient update, applying it directly to the weights as originally intended. This results in more stable training and improved model generalization, which is particularly crucial when dealing with the noisy gradients that can arise from hard-negative mining strategies. This ensures a more stable and effective regularization, which is important for preventing the model from overfitting, particularly given the noisy gradients that can arise from hard-negative mining [36].

The optimizer was paired with the `CosineAnnealingWarmRestarts` learning rate schedule. This scheduler, also introduced by Loshchilov and Hutter [37], was chosen to address the challenge of navigating the complex, non-convex loss landscapes that can arise from triplet loss, where an optimizer can become trapped in sharp, suboptimal local minima. Two powerful concepts drive this schedule: cosine annealing, which smoothly decays the learning rate following a cosine curve, and “warm restarts,” where the rate is periodically reset to its initial maximum [21]. By allowing the optimizer to escape suboptimal local minima that it may have settled into and explore other regions of the complex loss landscape, the scheduler increases the likelihood of finding a higher-quality solution [37].

Finally, the data loading and batching strategy was intrinsically linked to the fine-tuning objective. An inescapable constraint of the triplet loss functions is that the training data must contain a minimum of two examples for each class label to form a valid (anchor, positive) pair [28]. To mitigate the risk of creating mini-batches that fail this constraint through

simple random sampling, this research leveraged the `GroupByLabelBatchSampler` from the `sentence-transformers` library. This sampler ensures that each mini-batch is constructed by grouping samples with the same label, thereby guaranteeing every batch contains the necessary class diversity to form valid and informative triplets for all anchors [29].

Candidate Classification Algorithms

To determine the most effective method for classifying the generated pairwise feature vectors, a broad suite of machine learning algorithms was evaluated. The process began with a comprehensive initial evaluation of eight different models to understand which algorithmic families were best suited to the data. This initial suite included:

- **Linear Models (Logistic Regression, Ridge, Lasso):** To establish a baseline and test for linear separability.
- **Instance-Based Model (k-Nearest Neighbors):** To probe the local structure and clustering of the feature space.
- **Kernel-Based Model (Support Vector Machine):** To test for complex, non-linear decision boundaries.
- **Ensemble Model (Random Forest):** For its robustness and ability to capture complex feature interactions.

- **Probabilistic Models (Linear/Quadratic Discriminant Analysis):** To test assumptions about the geometric distribution of the data.

Based on the preliminary results from this comprehensive evaluation, four models were selected for the final, in-depth analysis due to their consistently strong performance: K-Nearest Neighbors (KNN), Support Vector Machine (SVM), Random Forest (RF), and XGBoost. XGBoost, a gradient boosting implementation, was added at this stage to include a state-of-the-art boosting algorithm known for its high performance on structured data, ensuring the final comparison was robust and comprehensive.

3.4 Evaluation Framework and Metrics

A rigorous, multi-stage evaluation framework was designed to narrow down the optimal combination of embedding models, dimensionality reduction techniques, and classifiers, and to provide a comprehensive view of the entire pipeline’s performance. This section details the design of that framework, the metrics used to measure performance, and the statistical methods planned for the final analysis.

Multi-Stage Evaluation Design

The evaluation process was designed as a funnel, beginning with a broad preliminary review to efficiently narrow the field of candidates, followed by a more focused and rigorous analysis on the larger, more comprehensive PPM Corpus.

Stage 1: Initial LLM Feasibility and Benchmark

The process began with a broad analysis to assess the feasibility of using a Large Language Model for direct classification and to contextualize its performance. This stage, conducted on the Initial Dataset, involved two parts:

1. **Feasibility Study:** The primary evaluation tested Google’s Gemini Pro v1.0 across different input data types (full raw text vs. extracted topics) and three classification schemes (binary, 3-way, and 4-way) to understand its capabilities and limitations.
2. **Comparative Benchmark:** To contextualize the results, a comparative analysis was conducted against a suite of other prominent open-source LLMs using the same binary classification task.

Stage 2: Feature Vector Validation (Ablation Study)

A critical ablation study was designed to validate the efficacy of the novel composite distance vector, Δ_c . This involved training the full suite of initial classifiers on two different versions of the feature set derived from the Initial Dataset: one using the complete composite vector and one using an ablated vector containing only the element-wise differences. By comparing the performance across these two conditions, we aimed to isolate and measure the contribution of the global cosine similarity feature.

Stage 3: Evaluation During Fine-Tuning

With the larger, more robust PPM dataset, the most promising non-proprietary embedding models were fine-tuned to specialize them for the course equivalency task. To monitor the quality of the learned embeddings during this process, the `BinaryClassificationEvaluator` from the `sentence-transformers` library was employed. At the end of each training epoch, the evaluator was run on binary-labeled course pairs generated from the training set. The primary metric monitored was Average Precision based on cosine similarity, and the model checkpoint that achieved the highest score on these validation pairs was saved as the best model for that fine-tuning run.

Stage 4: Final Downstream Classifier Evaluation

The final and definitive evaluation was conducted on the held-out test portion of the PPM dataset. This stage used the feature vectors generated from the best-performing embedding models (both the original off-the-shelf versions and the newly fine-tuned versions). These feature sets were then used to train and evaluate the final selection of high-performing classifiers: KNN, Random Forest, SVM, and XGBoost. This ensures an unbiased assessment of the complete pipeline's ability to generalize to new, unseen data. To find the optimal hyperparameters for each classifier, an exhaustive, brute-force hyperparameter grid search was conducted with 5-fold cross-validation.

Performance Metrics

To facilitate a comprehensive and multi-faceted analysis, a suite of standard evaluation metrics was used to assess the various pipeline configurations across two critical dimensions: classification efficacy and computational efficiency.

Classification Efficacy

The core assessment of classification performance is based on a standard suite of metrics derived from the confusion matrix, which tabulates the counts of True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN). While Accuracy, defined as $\frac{TP+TN}{TP+TN+FP+FN}$, provides a general overview of correctness, it can be insufficient for capturing the nuances of a classifier's behavior. Therefore, this research also evaluates:

- **Precision:** Calculated as $\frac{TP}{TP+FP}$, this metric measures a model's exactness. High precision indicates that when the model predicts an equivalency, it is likely to be correct.
- **Recall:** Calculated as $\frac{TP}{TP+FN}$, this metric measures a model's completeness. High recall indicates that the model is effective at identifying the full set of all true equivalencies.

The F_1 -Score, the harmonic mean of Precision and Recall ($2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$), is used as the primary metric for reporting and comparing the final classification performance. The F_1 -Score provides a single, robust value that balances the trade-off between precision and

recall, making it an ideal metric for this task where both avoiding false equivalencies and identifying all true ones are important.

Efficiency Metrics

Beyond classification efficacy, the practical viability of each pipeline was assessed by measuring its computational efficiency. Both Training Time and Inference Time were captured during the hyperparameter tuning process using the detailed statistics provided by `scikit-learn`'s `GridSearchCV`⁴ utility. Training time reflects the resources required to fit a model configuration, offering insight into the cost of experimentation. Inference time, reported by `GridSearchCV` as `score_time`, measures the time required for a trained model to make predictions on new data. For the use case of this research, inference time is considered the more critical efficiency metric, as it directly impacts the system's real-world responsiveness and scalability in a production environment. While memory and space requirements are also important, they are considered a less critical bottleneck for this application given the relative abundance of RAM and cache in modern hardware.

Statistical Analysis Plan

To determine if the observed differences in mean performance between models were statistically significant, a formal hypothesis testing procedure was conducted for the analysis of the held-out test data. A one-way Analysis of Variance (ANOVA) was carried out to first test

⁴SciKit Learn's `GridSearchCV` class can be found at: https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html

the null hypothesis that all four finalist classifiers have equal mean F_1 -scores. A statistically significant result would justify a post-hoc analysis to identify which specific pairs of classifiers differ. Prior to this, the assumption of homogeneity of variances would be checked using Levene’s Test (with the Brown-Forsythe correction) [8]. This test is critical, as a violation of this assumption would make standard post-hoc tests like Tukey’s HSD inappropriate. In the event of unequal variances, the Games-Howell post-hoc test was selected for its robustness, as it does not assume equal variances [22]. This ensures a methodologically rigorous comparison.

3.5 Misclassification Analysis Methodology

While the quantitative evaluation presented in the preceding sections establishes the performance of the pipeline based on aggregate metrics like the F_1 -score, a purely numerical analysis is insufficient for a complete understanding of the system’s behavior. High-level metrics are essential for benchmarking and model selection, but they obscure the underlying nature of a model’s failures. They quantify what the performance is but fail to explain why and how errors occur. Relying on such metrics alone can be misleading, as they may overestimate a model’s robustness while hiding important failure modes [23].

A critical step in the development of reliable and trustworthy AI systems is a thorough error analysis that moves beyond aggregate scores to diagnose specific weaknesses. This section, therefore, outlines the methodology for a qualitative diagnosis of the system. The

objective is to perform an error analysis on the misclassifications produced by the embedding models to understand the root causes of their failures. This investigation follows a two-pronged methodology.

Shared Misclassifications Across Models

An major step in pinpointing the origin of model failures is to differentiate between errors that are specific to individual models and those that are shared among them. If misclassifications are largely unique to each model, the errors are likely attributable to model-specific factors, such as architectural differences or suboptimal fine-tuning. Conversely, a significant overlap in misclassified pairs across multiple, diverse models would suggest the presence of failures that stem not from the models but from the data itself. For example, such errors may arise from annotation artifacts, inherent ambiguity in the source text, or insufficient information to support a clear classification, and they represent a fundamental challenge for any model applied to the dataset.

The first prong of the analysis, therefore, examines these error patterns by analyzing the overlap of misclassified course pairs among the different embedding models. This approach, using visualizations such as Venn diagrams and intersection bar charts, helps distinguish between model-specific weaknesses and challenges that are inherent to the dataset itself.

Qualitative Case-Based Review

The second prong of the analysis is a granular, qualitative review of specific false positive and false negative examples drawn from the held-out test and cross-validation reports. This manual, case-based review is a cornerstone of effective error analysis, enabling the identification of specific artifacts and conceptual flaws that quantitative metrics cannot reveal. This is particularly meaningful for understanding failures in complex semantic tasks. For example, it allows for the investigation of challenges in short-text semantic similarity, where a lack of rich context inherently increases ambiguity and the likelihood of spurious matches [4].

The overarching goal of this two-pronged methodology is to generate a structured report of the primary error types. This includes identifying and categorizing the root causes of both False Positives (e.g., high topical overlap without true equivalence) and False Negatives (e.g., semantic divergence in descriptions or data quality issues). This diagnostic approach is designed to inform future improvements to the system by revealing where the pipeline is most likely to fail and why.

Chapter 4

Experimental Setup and Results

This chapter presents the empirical results that validate the decoupled, deep metric learning framework proposed in Chapter 3. The central objective is to exhaustively assess the performance of each component of the pipeline—from the choice of embedding model and the impact of fine-tuning to the effects of dimensionality reduction and the selection of a downstream classifier—to identify the optimal configuration for both classification accuracy and computational efficiency.

The analysis is structured to first establish the validity of the core feature engineering approach before proceeding through the primary evaluation sequence. The chapter begins by presenting an ablation study to validate the efficacy of the novel composite distance vector (Δ_c), the cornerstone of the feature representation. With the feature vector’s design validated, the investigation then establishes a baseline by assessing off-the-shelf models, quantifies the performance gains achieved through fine-tuning, and analyzes the trade-offs

between accuracy and efficiency. The chapter culminates in a definitive comparative analysis on the held-out test data, synthesizing all prior findings to identify the single best-performing pipeline configuration. We begin by outlining the experimental setup that forms the foundation for all subsequent results.

4.1 Experimental Environment & Datasets

Computational Environment

The research presented in this thesis was conducted using a hybrid computational environment, leveraging both cloud-based services for initial language model evaluations and a powerful on-premise high-performance computing (HPC) cluster for the primary, computationally intensive experiments. The initial direct classification tasks were performed using Google’s Gemini v1.0, a proprietary cloud-based Large Language Model. All subsequent stages, including model fine-tuning and downstream classifier evaluations, were executed on San Francisco State University’s “POLARIS” High Performance Compute Cluster. The POLARIS cluster runs on Rocky Linux 8.9 with the Slurm Workload Manager. GPU-intensive deep learning tasks were performed on a node equipped with four NVIDIA A100 GPUs, while extensive hyperparameter grid searches for traditional machine learning classifiers were run on a CPU cluster with AMD EPYC 9534 CPUs. The entire experimental pipeline was implemented in Python, with the core deep learning components built using PyTorch and the

Hugging Face ecosystem and the classical machine learning experiments conducted using `scikit-learn` and `XGBoost`.

Datasets

As detailed in Chapter 3, this research utilized two distinct datasets at different stages of the evaluation. A smaller **Initial Dataset**, manually curated from the ASSIST repository, was used for prototyping and the preliminary screening of models and classifiers. The definitive experiments were conducted on the larger **PPM Corpus**, provided by the Program Pathways Mapper (PPM). This corpus was used for the definitive fine-tuning and final pipeline evaluation, with its held-out test set reserved for an unbiased assessment of the optimized pipelines to ensure methodological rigor. Table 4.1 provides a summary of the key characteristics of both datasets.

Table 4.1: Summary of Datasets Used in Evaluation

Characteristic	Initial Dataset	PPM Corpus
Source	Manually curated via ASSIST	Program Pathways Mapper (PPM)
Purpose	Preliminary screening, prototyping, and initial classifier evaluation	Definitive fine-tuning and final pipeline evaluation
Ground Truth	Established articulation agreements	Course Identification Number (C-ID)
Final Size	400 course pairs	9192 course pairs
Partitioning	Stratified random sample	Stratified 50/50 train/test split

Evaluation Metrics

To facilitate a multi-faceted analysis, a standard suite of metrics was used to assess both classification efficacy and computational efficiency. The core assessment of classification performance was measured using the F_1 -Score, the harmonic mean of Precision and Recall. This metric was chosen because it provides a single, robust value that balances the trade-off between avoiding false equivalencies (precision) and identifying all true ones (recall), making it ideal for this task. For assessing practical viability, both Training Time and Inference Time were captured. Inference time is considered the more meaningful efficiency metric for this research, as it directly impacts the system’s real-world responsiveness and scalability in a production environment.

4.2 Baseline Performance: Direct LLM Classification

The initial phase of this research sought to establish a performance baseline by evaluating the feasibility of using a Large Language Model for end-to-end course equivalency classification, a methodology detailed in Section 3.1. The experiments were conducted on the Initial Dataset, leveraging Google’s Gemini Pro v1.0 as the reasoning engine. The results were encouraging, with the model achieving a peak accuracy of 90.5% when using full, unprocessed raw text as input. This approach consistently yielded superior results compared to using only extracted structured topics. This performance gap is likely attributable to the challenges inherent in the data extraction step itself, which proved to be difficult and

Table 4.2: Performance Summary of Direct LLM Classification

Classification		Accuracy	F1-Score*	Precision*	Recall*
Input Data	Task				
Raw Text	Binary	0.9050	0.8973	0.9765	0.8300
	3-way	0.8750	0.8877	0.9818	0.8100
	4-way	0.8100	0.8596	0.9808	0.7650
Topics	Binary	0.8100	0.7654	1.0000	0.6200
	3-way	0.7775	0.7795	0.9847	0.6450
	4-way	0.7225	0.7531	0.9839	0.6100

* F1-Score, Precision, and Recall are reported for the positive class (“Equivalent”).

highly sensitive to prompt wording—a finding that aligns with previous studies on prompt engineering [57, 20, 12, 13, 24].

An analysis of the confusion matrices, presented in Figure 4.1, reveals a consistent and telling behavioral pattern: the model exhibits a strong conservative bias. Across all scenarios, it was far more likely to misclassify a truly equivalent course pair as not equivalent (a false negative) than it was to incorrectly approve a non-equivalent pair (a false positive), suggesting the LLM operates with a high internal threshold for declaring equivalency. Furthermore, the experiments that introduced “unsure” and “insufficient data” categories demonstrated the model’s ability to effectively isolate ambiguous cases that would require manual review. A comprehensive summary of the performance metrics is provided in Table 4.2, and a comparative benchmark against other prominent LLMs is shown in Table 4.3.

This initial study confirmed that while direct LLM classification can achieve high accuracy, it has fundamental limitations, including high computational cost, lack of a quantifiable

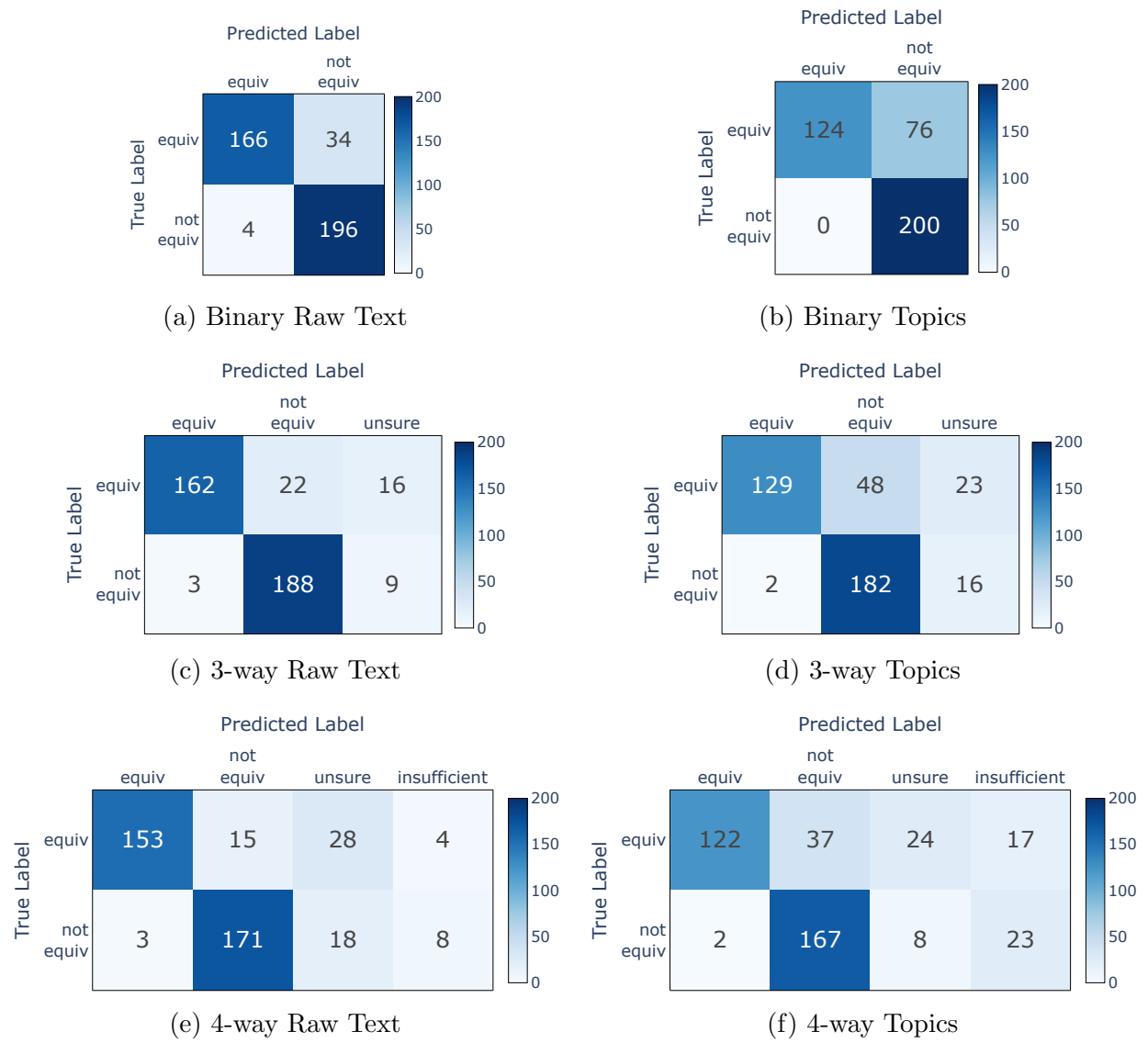


Figure 4.1: LLM Classification Confusion Matrices

Table 4.3: Model Specifications and Performance

Model Name	Parameters*	Context	Support	Accuracy	Precision	Recall	F ₁ -score
		Length					
Google Gemini Pro 1.0	Unknown	32,768	400	0.9050	0.9765	0.8300	0.8973
Meta Llama 3.1 8B Instruct	8	128,000	208 (52%)	0.6250	1.0000	0.3500	0.5185
Microsoft Phi 3 Medium Instruct	14	128,000	400	0.7100	1.0000	0.4200	0.5915
Google Gemma 2 27b	27.2	8,000	N/A	N/A	N/A	N/A	N/A
Meta Llama 3.1 70B Instruct	70	128,000	400	0.7350	1.0000	0.4700	0.6395
Qwen 2 72B Instruct	72.7	131,072	400	0.7650	1.0000	0.5300	0.6928
Anthracite Magnum v1 72B	72.7	32,768	400	0.8525	1.0000	0.7050	0.8270
CalmeRys 78B Orpo v0.1	78	32,768	400	0.7175	1.0000	0.4350	0.6063
Mixtral 8x22B Instruct v0.1	141	65,536	400	0.6450	1.0000	0.2900	0.4496
Meta Llama 3.1 405B Instruct**	405	128,000	400	0.7775	1.0000	0.5550	0.7138

*in Billions

**INT4 Quantized Model

similarity score, sensitivity to input quality, and an opaque, “black box” nature. These challenges validated the decision to pivot and motivated the development of the more robust, decoupled pipeline that is the primary focus of this thesis.

4.3 Core Component Validation: The Composite

Distance Vector

A foundational contribution of this research is the composite distance vector, Δ_c , introduced in Section 3.2. This vector was designed to provide a richer feature set for downstream classifiers by combining granular, dimension-specific differences with a holistic, global cosine similarity metric. To validate this design, an ablation study was conducted to quantify the impact of the global cosine similarity term on classifier performance whose results are summarized in Figure 4.2. Classifiers were trained on embeddings from each of the finalist embedding models (as discussed in Section 3.3), using both the complete vector (Δ_c) and an

ablated version containing only the local, element-wise differences (Δ_l).

		PCA						TSNE						PaCMAP					
4 Dimensions		bge		gist		nve		bge		gist		nve		bge		gist		nve	
		no cos	cos	no cos	cos	no cos	cos	no cos	cos	no cos	cos	no cos	cos	no cos	cos	no cos	cos	no cos	cos
		LR: 0.5843	0.9486	0.6046	0.9789	0.6222	0.9757	0.5540	0.8593	0.5711	0.9365	0.5495	0.9270	0.6406	0.9637	0.5914	0.9855	0.6509	0.9835
	Ridge:	0.6093	0.9428	0.6604	0.9729	0.6374	0.9691	0.5540	0.8554	0.5713	0.9275	0.5497	0.9132	0.6744	0.9241	0.5980	0.9830	0.6627	0.9835
	Lasso:	0.6167	0.9488	0.6824	0.9796	0.6715	0.9758	0.5612	0.8598	0.5736	0.9376	0.5589	0.9271	0.7647	0.9661	0.7456	0.9885	0.7858	0.9887
	KNN:	0.9718	0.9775	0.9893	0.9908	0.9910	0.9915	0.9155	0.9162	0.9294	0.9285	0.9538	0.9500	0.9802	0.9818	0.9942	0.9951	0.9944	0.9957
	SVM:	0.9722	0.9772	0.9893	0.9908	0.9905	0.9915	0.9238	0.9372	0.9331	0.9526	0.9597	0.9643	0.9789	0.9791	0.9928	0.9940	0.9942	0.9950
	RF:	0.9694	0.9747	0.9886	0.9883	0.9888	0.9919	0.9170	0.9417	0.9232	0.9517	0.9507	0.9620	0.9791	0.9813	0.9933	0.9950	0.9927	0.9965
	LDA:	0.5844	0.9423	0.6049	0.9723	0.6244	0.9686	0.5539	0.8553	0.5710	0.9269	0.5495	0.9128	0.6412	0.9239	0.5943	0.9830	0.6518	0.9835
	QDA:	0.9518	0.9544	0.9636	0.9779	0.9687	0.9781	0.8965	0.8882	0.9252	0.9420	0.9424	0.9479	0.9628	0.9635	0.9874	0.9906	0.9923	0.9925
7 Dimensions	LR:	0.6185	0.9504	0.6189	0.9736	0.6302	0.9728	0.5785	0.8385	0.5733	0.9273	0.5567	0.8607	0.5717	0.9597	0.5844	0.9859	0.6412	0.9820
	Ridge:	0.6328	0.9471	0.6569	0.9711	0.6446	0.9677	0.5792	0.8387	0.5738	0.9139	0.5567	0.8574	0.6346	0.9264	0.6000	0.9849	0.6725	0.9811
	Lasso:	0.6334	0.9495	0.6801	0.9731	0.6712	0.9723	0.5821	0.8388	0.5831	0.9275	0.5687	0.8612	0.7824	0.9674	0.7931	0.9890	0.7930	0.9855
	KNN:	0.9775	0.9839	0.9902	0.9949	0.9937	0.9935	0.9365	0.9344	0.9407	0.9456	0.9555	0.9542	0.9818	0.9840	0.9933	0.9947	0.9959	0.9963
	SVM:	0.9825	0.9856	0.9923	0.9941	0.9951	0.9941	0.9420	0.9563	0.9496	0.9713	0.9680	0.9793	0.9823	0.9835	0.9938	0.9948	0.9961	0.9964
	RF:	0.9732	0.9790	0.9894	0.9902	0.9901	0.9923	0.9343	0.9547	0.9371	0.9621	0.9559	0.9747	0.9820	0.9862	0.9945	0.9955	0.9956	0.9965
	LDA:	0.6189	0.9469	0.6192	0.9707	0.6315	0.9670	0.5792	0.8371	0.5734	0.9133	0.5567	0.8574	0.5725	0.9264	0.5842	0.9849	0.6424	0.9811
	QDA:	0.9629	0.9635	0.9761	0.9793	0.9754	0.9816	0.9222	0.9091	0.9325	0.9455	0.9509	0.9424	0.9746	0.9750	0.9925	0.9922	0.9932	0.9938
		No Dimesion Reduction																	
		bge		gist		nve													
		no cos	cos	no cos	cos	no cos	cos												
All Dimensions	LR:	0.6859	0.9013	0.6845	0.9488	0.6865	0.9476												
	Ridge:	0.6901	0.9066	0.6885	0.9454	0.6928	0.9447												
	Lasso:	0.6826	0.9093	0.6821	0.9513	0.6812	0.9470												
	KNN:	0.9818	0.9833	0.9893	0.9901	0.9893	0.9910												
	SVM:	0.9988	0.9983	0.9993	0.9986	0.9993	0.9994												
	RF:	0.9830	0.9807	0.9890	0.9866	0.9928	0.9932												
	LDA:	0.6876	0.9068	0.6876	0.9446	0.6876	0.9435												
	QDA:	0.0410	0.0487	0.0005	0.0005	0.0000	0.1333												

Figure 4.2: A comparison of composite distance vs. elementwise-distance only.

The results revealed a stark dichotomy between the behavior of linear and non-linear models. The performance of linear models, such as Logistic Regression, showed a dramatic improvement with the inclusion of the cosine similarity term. For example, the F_1 -score for Logistic Regression surged from 0.686 to 0.901 with the addition of the single global feature. This demonstrates that these simpler models, which are constrained to learning linear decision boundaries, rely on this explicit global feature and are unable to derive the relationship from the local difference vector alone. In contrast, the performance of more

complex, non-linear models (e.g., KNN, SVM, and Random Forest) was not noticeably affected. Their exceptionally high performance remained stable with or without the cosine similarity term, suggesting these more powerful models are capable of inferring the global relationship directly from the high-dimensional local difference vector.

The results of this ablation study are conclusive regarding the utility of the composite distance vector, Δ_c . For linear models, its inclusion is vital, providing a substantial boost in performance. While the more powerful non-linear models did not derive a consequential benefit, their performance was not negatively impacted. Therefore, to maintain a consistent and robust feature engineering pipeline, the composite distance vector was retained as the standard feature representation for all subsequent experiments. The exploration into alternative composite vectors that could potentially yield improvements for non-linear models is an avenue for future research.

4.4 Domain-Specific Adaptation: Embedding Model

Fine-Tuning

This section details the empirical investigation into enhancing a pre-trained embedding model by fine-tuning it on the domain-specific PPM Corpus. This process, a form of deep metric learning, aims to restructure the embedding space such that geometric distance directly corresponds to semantic similarity. Such an approach is particularly relevant for sce-

narios with high intra-class variance and low inter-class variance, a common characteristic of specialized domains where fine distinctions are paramount [43].

Fine-Tuning Experiment Summary

To identify the optimal fine-tuning configuration, an evaluation was designed to test the impact of different loss functions and learning rate schedules on the BAAI/bge-small-en-v1.5 (BGE) base model. A matrix of twelve experimental configurations was executed, crossing the four triplet mining loss functions and three learning rate schedulers whose theoretical underpinnings were detailed in Section 3.3. The training was conducted on the POLARIS HPC cluster using a distributed data-parallel strategy. To ensure efficient and robust training, the `GroupByLabelBatchSampler` was employed for effective triplet mining, and model performance was monitored at the end of each epoch using the F_1 -score on a binary classification task. This was a deliberate choice to evaluate the model on its ultimate downstream application rather than the triplet loss objective itself, providing a more honest assessment of its ability to generalize.

Empirical Results

The examination of the twelve fine-tuning configurations produced a rich set of performance data. Figure 4.3 visualizes the validation F_1 -score over the training process, providing a qualitative comparison of the learning dynamics for each configuration. For a more precise

and formal comparison, the exact peak validation F_1 -scores achieved by each configuration are summarized in Table 4.4.

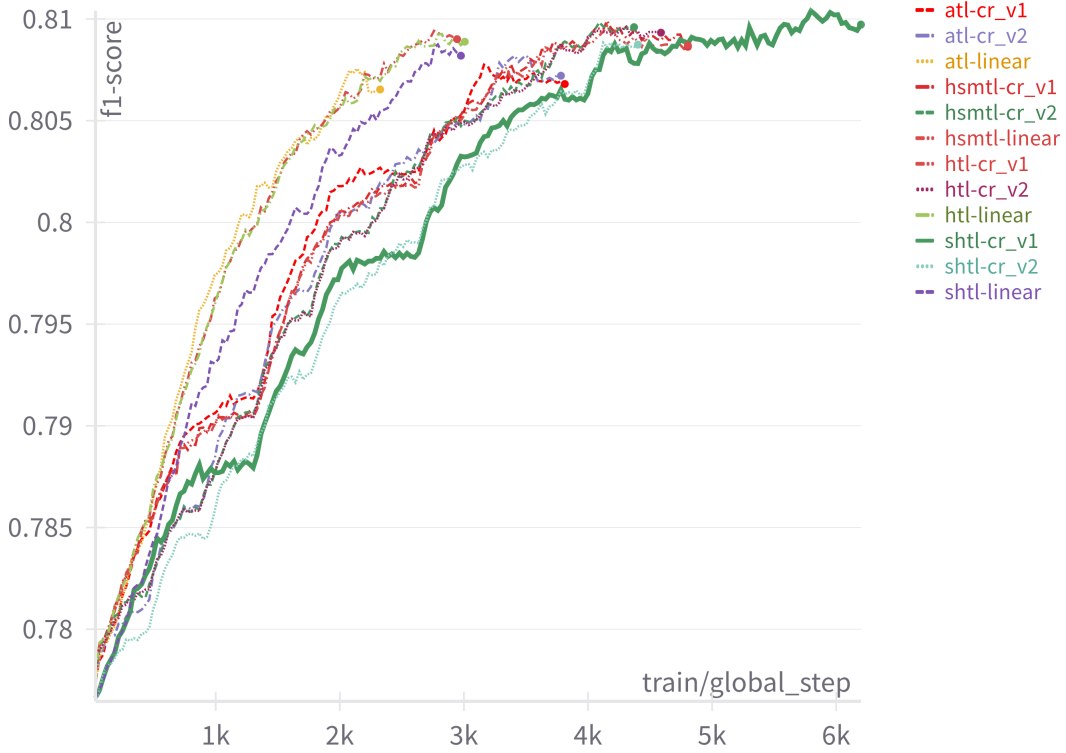
Figure 4.3: Validation F_1 -Score

Table 4.4: Peak Validation F1-Score of Fine-Tuning Configurations on BGE Model

Loss Function	Linear Scheduler	Cosine Restarts v1	Cosine Restarts v2
BatchAllTripletLoss (atl)	0.8076	0.8078	0.8082
BatchHardSoftMarginTripletLoss (hsmatl)	0.8094	0.8099	0.8098
BatchHardTripletLoss (htl)	0.8093	0.8097	0.8096
BatchSemiHardTripletLoss (shtl)	0.8088	0.8104	0.8088

The quantitative data reveals a nuanced outcome. While aggressive hard-negative mining strategies such as `BatchHardTripletLoss` and `BatchHardSoftMarginTripletLoss` consistently provided a high performance floor, the single best-performing configuration was

the combination of `BatchSemiHardTripletLoss` with the `cr_v1` cosine annealing schedule, which achieved the highest peak validation F_1 -score of 0.8104. This result suggests that a more moderate mining strategy, when paired with a patient and robust optimization schedule, can find a slightly better optimum for this specific task.

Validation and Selection of the Optimal Embedding Model

The superior performance of the winning configuration arises from a synergistic framework: a stable learning objective from semi-hard mining, a robust exploration strategy from the `CosineAnnealingWarmRestarts` scheduler that allows the optimizer to escape local minima, and effective regularization from the AdamW optimizer [36, 26, 56, 37]. To definitively validate the benefit of this fine-tuning process, the performance of the resulting model (BGE-ft) was compared against the suite of off-the-shelf models using the held-out test portion of the PPM Corpus. The descriptive statistics in Table 4.5 show that the fine-tuned BGE-ft model not only achieved the highest mean F_1 test score ($\mu = 0.9787$) but also exhibited the lowest standard deviation ($\sigma = 0.0043$), indicating it is both more accurate and more consistent on unseen data.

Table 4.5: Descriptive Statistics for F1-Scores by Embedding Model (Test Data)

Embedding Model	count	mean	std	min	25%	50%	75%	max
BGE	64	0.967681	0.015185	0.900304	0.957357	0.973832	0.978318	0.984612
BGE-ft	63	0.978746	0.004320	0.968707	0.976240	0.979922	0.981683	0.987702
GIST	63	0.971851	0.012516	0.913939	0.965367	0.977210	0.981077	0.984083
NVE	64	0.970530	0.013705	0.937957	0.960745	0.975147	0.982084	0.985258
SFR	64	0.972254	0.014617	0.922104	0.963553	0.978764	0.982926	0.986549

To determine if these observed performance differences were statistically significant, a formal hypothesis testing procedure was conducted. First, Levene’s test was performed to check the assumption of homogeneity of variances across the embedding model groups. As shown in Table 4.6a, the result was statistically significant ($p < 0.001$), indicating that the variances were unequal and justifying the use of a Welch’s ANOVA and the more robust Games-Howell post-hoc test. A one-way Welch’s ANOVA subsequently confirmed that a statistically significant difference exists between the model groups ($F(4, 141.9) = 15.9881, p < 0.001$), as detailed in Table 4.6b.

Table 4.6: Statistical Tests for F1-Scores by Embedding Model (Test Data)

(a) Levene’s Test for Homogeneity of Variances

Statistic (W)	df1	df2	p-value
7.0286	4	313	< 0.001

(b) Welch’s ANOVA

F-statistic	df1	df2	p-value
15.9881	4	141.90	< 0.001

The Games-Howell post-hoc test (Table 4.7) provided detailed pairwise comparisons, revealing that the BGE-ft model demonstrated a statistically significant improvement over its base model, BGE, with a mean F_1 -score increase of 0.0111 ($p < 0.001$). Notably, BGE-ft outperformed all other evaluated models by a considerable amount, including those that were orders of magnitude larger, such as NVE and NVE. This rigorous evaluation leads to a clear and unambiguous conclusion. The single best model for this task is the

BAAI/bge-small-en-v1.5 model fine-tuned using `BatchSemiHardTripletLoss` and a `CosineAnnealingWarmRestarts` learning rate schedule. This model, was proven to be statistically superior to all off-the-shelf models evaluated in this study and is carried forward as the primary fine-tuned embedding model for the comprehensive classifier evaluation detailed in the next section.

Table 4.7: Games-Howell Post-Hoc Test for Classifiers[†]

Group 1	Group 2	Mean	p-val*	95% CI	95% CI	Reject
		Diff.		Lower	Upper	
KNN	RF	-0.0007	0.9923	-0.0068	0.0054	False
KNN	SVM	0.0051	0.0577	-0.0001	0.0102	False
KNN	XGB	-0.0001	0.9999	-0.0058	0.0056	False
RF	SVM	0.0057	0.0178	0.0007	0.0107	True
RF	XGB	0.0005	0.9944	-0.0050	0.0061	False
SVM	XGB	-0.0052	0.0160	-0.0097	-0.0007	True

[†] Across all embedding models and all tests

* Adjusted p -value using Tukey’s method

4.5 Downstream Classifier Performance

This section presents the definitive evaluation of the downstream classifiers, representing the final component of the proposed framework. The objective is to identify the optimal classification model by balancing predictive accuracy against computational efficiency. The evaluation proceeds from a broad preliminary review to a focused analysis of four finalist classifiers on the PPM Corpus, culminating in a statistical comparison on held-out test data.

The Impact of Dimensionality Reduction

Before evaluating the final classifiers, it is crucial to analyze the impact of the dimensionality reduction techniques detailed in Section 3.2. The high-dimensional vectors produced by embedding models can pose challenges for some classifiers, and reducing dimensionality can potentially improve performance and efficiency. A comprehensive statistical analysis was performed on the test data to compare the efficacy of using the original embeddings versus those transformed by PCA, UMAP, and PaCMAP, as well as to compare different target dimensionalities.

The results, summarized in Tables 4.8 through 4.11, reveal a clear and pronounced finding: dimensionality reduction is not just a computational optimization but a key step for improving classification accuracy. The PaCMAP reducer, in particular, stands out, achieving a statistically significant improvement over all other methods, including using no reduction at all ($p < 0.05$ for all comparisons). Nevertheless, a particular item of note is that BGE-ft performed the best while using PCA. Similarly, when analyzing the target number of dimensions, reducing the feature space to a size that explains 70% or 80% of the original variance yields a statistically significant performance boost over using the original embeddings or reducing to a very small, fixed number of dimensions like 5. These results strongly suggest that dimensionality reduction techniques, particularly PaCMAP and PCA, effectively filter noise and distill the most discriminative signals from the high-dimensional embeddings, leading to a more robust feature set for the downstream classifiers. All subsequent classifier

evaluations, therefore, incorporate these optimized, dimensionally-reduced feature vectors.

Table 4.8: Descriptive Statistics for F1-Scores by Dimensionality Reducer (Test Data)

Dim. Reducer	count	mean	std	min	25%	50%	75%	max
None	20	0.962981	0.024292	0.900304	0.961691	0.972235	0.976458	0.983663
PaCMAP	98	0.981104	0.001835	0.977172	0.979452	0.981442	0.982431	0.984393
PCA	100	0.971991	0.012743	0.937957	0.962466	0.977626	0.982058	0.987702
UMAP	100	0.965505	0.011596	0.938061	0.956572	0.968961	0.974924	0.982183

Table 4.9: Games-Howell Post-Hoc Test for Dimensionality Reducers (Test Data)

Group 1	Group 2	Mean	p-adj	95% CI		Reject
		Diff.		Lower	Upper	
None	PaCMAP	0.0181	0.0167	0.0028	0.0334	True
None	PCA	0.0090	0.3920	-0.0065	0.0246	False
None	UMAP	0.0025	0.9680	-0.0130	0.0180	False
PaCMAP	PCA	-0.0091	0.001	-0.0125	-0.0058	True
PaCMAP	UMAP	-0.0156	0.001	-0.0187	-0.0125	True
PCA	UMAP	-0.0065	0.0013	-0.0110	-0.0020	True

Table 4.10: Descriptive Statistics for F1-Scores by Number of Dimensions (Test Data)

# Dimensions	count	mean	std	min	25%	50%	75%	max
5	60	0.966575	0.014254	0.937957	0.955814	0.969970	0.980417	0.983850
7	59	0.970692	0.009840	0.950073	0.962735	0.972467	0.980017	0.983124
70%	60	0.975803	0.009528	0.944029	0.973550	0.979290	0.982021	0.985604
80%	60	0.976336	0.009952	0.938061	0.975156	0.979441	0.982494	0.986549
90%	59	0.974646	0.012442	0.939868	0.974690	0.979041	0.982269	0.987702
Original	20	0.962981	0.024292	0.900304	0.961691	0.972235	0.976458	0.983663

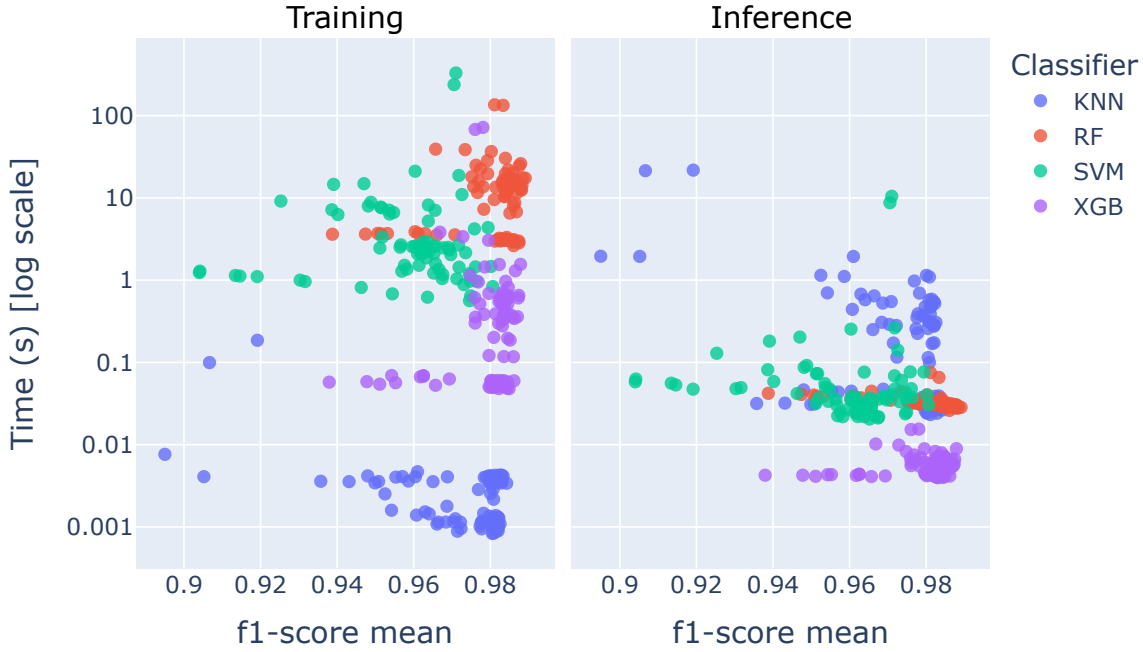
Efficacy vs. Efficiency Analysis

With the feature set optimized, the analysis next investigated the trade-off between model efficacy and computational efficiency for the four finalist classifiers: K-Nearest Neighbors

Table 4.11: Games-Howell Post-Hoc Test for Number of Dimensions (Test Data)

Group 1	Group 2	Mean	p-adj	95% CI		Reject
		Diff.		Lower	Upper	
5	7	0.0041	0.4473	-0.0024	0.0106	False
5	70%	0.0092	0.0009	0.0028	0.0157	True
5	80%	0.0098	0.0004	0.0032	0.0163	True
5	90%	0.0081	0.0162	0.0010	0.0152	True
5	Original	-0.0036	0.9878	-0.0214	0.0142	False
7	70%	0.0051	0.0527	-0.0000	0.0103	False
7	80%	0.0056	0.0277	0.0004	0.0109	True
7	90%	0.0040	0.3988	-0.0020	0.0099	False
7	Original	-0.0077	0.7370	-0.0252	0.0097	False
70%	80%	0.0005	0.9997	-0.0046	0.0057	False
70%	90%	-0.0012	0.9928	-0.0071	0.0047	False
70%	Original	-0.0128	0.2368	-0.0302	0.0046	False
80%	90%	-0.0017	0.9638	-0.0077	0.0043	False
80%	Original	-0.0134	0.2033	-0.0308	0.0041	False
90%	Original	-0.0117	0.3432	-0.0293	0.0060	False

(KNN), Support Vector Machine (SVM), Random Forest (RF), and XGBoost. Figure 4.4 visualizes the median F_1 -scores from the cross-validation grid search against both training and inference times. While all four models demonstrate the ability to achieve high accuracy, Random Forest and XGBoost emerge as the clear winners with respect to inference time. Their inference times are tightly clustered under 0.1 seconds—an order of magnitude faster and more predictable than the wider, more variable times of KNN and SVM. This reliability makes RF and XGBoost inherently more robust choices for a scalable, low-latency production system.

Figure 4.4: Classifier F_1 -Score vs. Training and Inference Time

Statistical Analysis of Classifier Performance on Test Data

The definitive comparison was conducted on the held-out test data from the PPM Corpus. As shown in the score distributions (Figure 4.5) and descriptive statistics (Table 4.12), all four classifiers demonstrate exceptionally high and stable performance across all embedding models. To provide a more comprehensive view of classifier performance beyond the F_1 -score, Receiver Operating Characteristic (ROC) curves were generated, as shown in Figure 4.6b. The Area Under the Curve (AUC) scores for all models are exceptionally high (all > 0.98), indicating excellent discrimination capability across all classification thresholds. The zoomed-in Figure 4.6a highlights the subtle but consistent superiority of the SVM model. Further visualization through confusion matrices (Figure 4.7) for the best-performing

hyperparameter configuration of each classifier paired with BGE-ft confirms their strong performance, with very few misclassifications overall. For more detailed ROC curves and confusion matrices for every embedding model, see Appendix B.

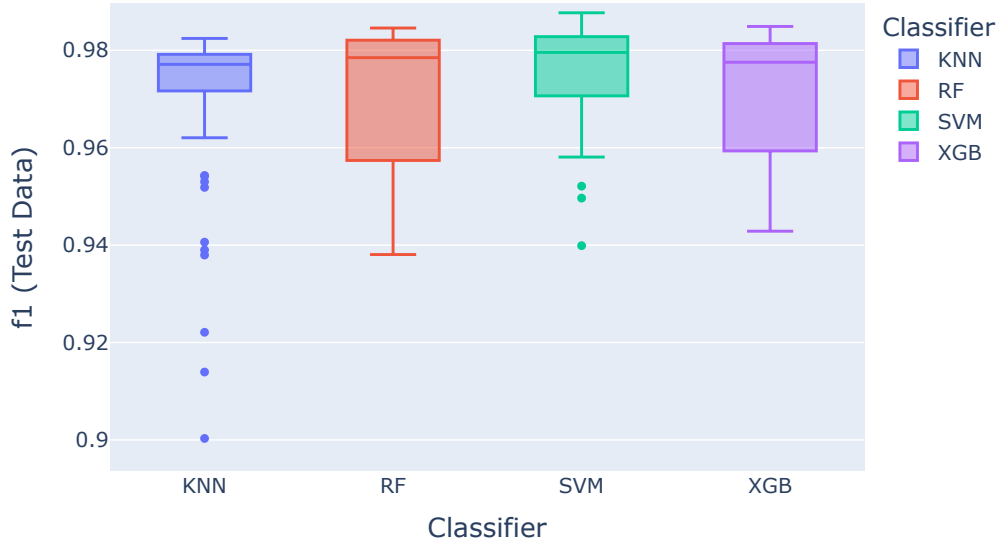


Figure 4.5: F_1 -Score Distribution of Finalist Classifiers Across All Embedding Models

Table 4.12: Classifier Descriptive Statistics on Test Data F_1 -Scores

Classifier	Count	Mean	Std	Min	25%	50%	75%	Max
KNN	80	0.971147	0.015208	0.900304	0.971809	0.977110	0.979110	0.982437
RF	80	0.970490	0.014473	0.938061	0.957396	0.978490	0.982073	0.984572
SVM	78	0.976210	0.009112	0.939885	0.970793	0.979567	0.982795	0.987702
XGB	80	0.971026	0.012316	0.942860	0.959382	0.977547	0.981376	0.984898

Notably, the Support Vector Machine (SVM) model achieved both the highest mean F_1 -score ($\mu = 0.9762$) and the lowest standard deviation ($\sigma = 0.0091$), indicating it is the most accurate and consistent performer. To determine if these observed differences were statistically significant, a one-way Welch’s ANOVA was performed, which confirmed a

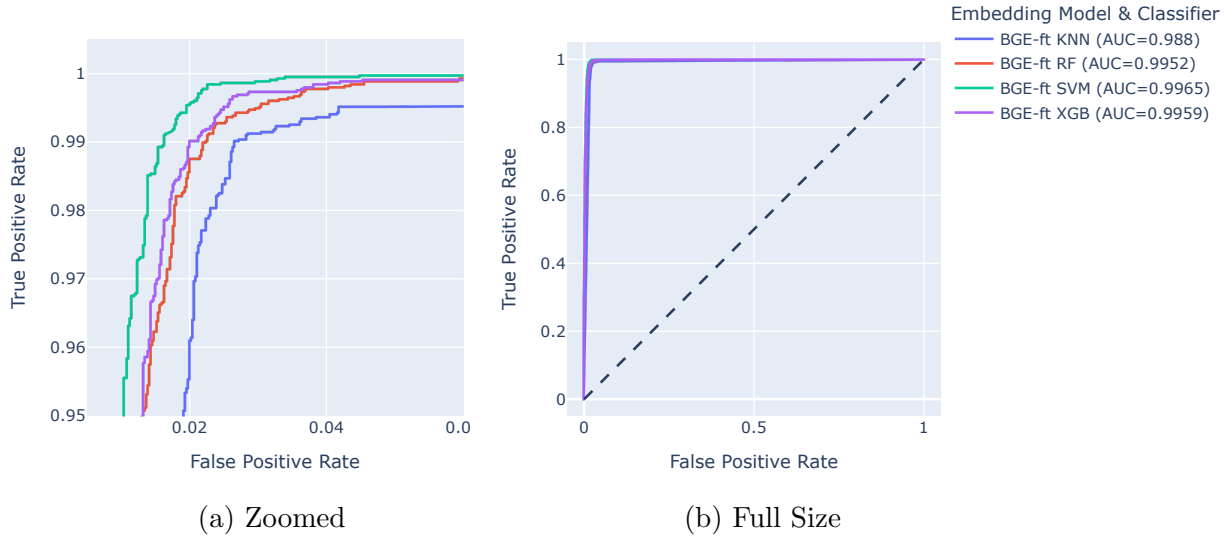


Figure 4.6: BGE-ft ROC Curves for Finalist Classifiers with Test Data

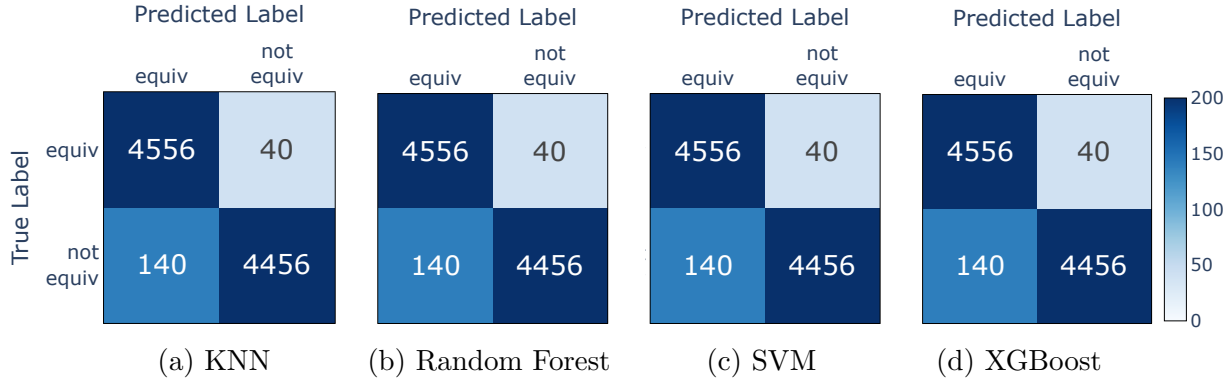


Figure 4.7: BGE-ft Confusion Matrices for Finalist Classifiers with Test Data

significant difference among the classifiers ($F(3, 171.4) = 5.0484, p = 0.0022$). A subsequent Games-Howell post-hoc test, justified due to unequal variances confirmed by Levene's Test ($p = 0.0387$), revealed that the SVM classifier performed statistically significantly better than both Random Forest (mean difference = 0.0057, $p = 0.0178$) and XGBoost (mean difference = 0.0052, $p = 0.016$). While all four are top-tier performers, the formal statistical analysis identifies SVM as the definitive winner in terms of raw accuracy. The full statistical

results are presented in Tables 4.12 and 4.13.

Table 4.13: Games-Howell Post-Hoc Test for Classifiers (Test Data)

Group 1	Group 2	Mean Diff.	p-adj	95% CI Lower	95% CI Upper	Reject
KNN	RF	-0.0007	0.9923	-0.0068	0.0054	False
KNN	SVM	0.0051	0.0577	-0.0001	0.0102	False
KNN	XGB	-0.0001	0.9999	-0.0058	0.0056	False
RF	SVM	0.0057	0.0178	0.0007	0.0107	True
RF	XGB	0.0005	0.9944	-0.0050	0.0061	False
SVM	XGB	-0.0052	0.0160	-0.0097	-0.0007	True

Final Classifier Selection

This multi-stage evaluation reveals a classic trade-off between peak performance and operational efficiency. The statistical superiority of SVM is clear; however, it corresponds to a mean F_1 -score improvement of only about 0.5% over the far more efficient tree-based models. The final choice is therefore context-dependent. Based on the comprehensive evidence, the following recommendations are made:

- **For Maximum Accuracy:** The Support Vector Machine (SVM) is the recommended classifier for any scenario where achieving the absolute highest accuracy and consistency is the paramount objective.
- **For Optimal Efficiency:** Random Forest and XGBoost are compelling and practical alternatives for applications where inference speed, low latency, and predictable per-

formance are essential for scalability. They provide nearly equivalent accuracy with demonstrably superior and more reliable computational efficiency.

4.6 Qualitative Diagnosis: Misclassification Analysis

While quantitative metrics are essential for benchmarking and model selection, they obscure the underlying nature of a model’s failures. Relying on aggregate scores alone can be misleading, as they may overestimate a model’s robustness while hiding serious failure modes [23]. This section, therefore, transitions from quantitative assessment to a qualitative diagnosis of the misclassifications produced by the embedding models to understand their root causes. The analysis examines shared misclassifications across models and conducts a granular, case-based review of false positives and false negatives.

Shared Misclassifications Across Models

To diagnose the source of model errors, it is first necessary to determine whether they are idiosyncratic to individual models or if they represent a common pattern of failure. A noteworthy overlap in misclassified pairs across multiple, diverse models would suggest the presence of errors that stem not from the models but from the data itself, such as inherent ambiguity or annotation artifacts. Individual course description texts that were most often misclassified are located in Appendix C.1.

The analysis of misclassifications on the held-out test data reveals a high degree of overlap,

providing strong evidence for the prevalence of a recurring issue found across multiple models. As illustrated in Figure 4.8, a total of 211 distinct course pairs were misclassified by every single embedding model evaluated, from the large, general-purpose models to the small, domain-specific BGE-ft. The intersection bar charts in Figure 4.9 reinforce this finding, showing high counts of shared errors across all model combinations. The interpretation is clear: a substantial portion of the model failures are not random but are a widespread problem affecting the course catalog corpus. These “hard” examples consistently challenge a range of semantic embedding models and may define the performance ceiling of any approach relying solely on course descriptions.

Qualitative Analysis of False Positives

A False Positive (FP) occurs when two non-equivalent courses are incorrectly classified as equivalent. This type of error carries meaningful real-world consequences, as it could lead a student to waste time and tuition on a non-transferable course. The analysis identified two primary causes:

- **Topical Overlap without True Equivalence:** This error arises when courses cover the same broad subject but differ in scope such as academic level or their position in a curricular sequence. The model correctly identifies high semantic similarity but fails to capture the nuanced distinctions that make the courses non-equivalent. An example is the pair of PHYS-4D and PHYS-4A from Foothill College, which are sequential

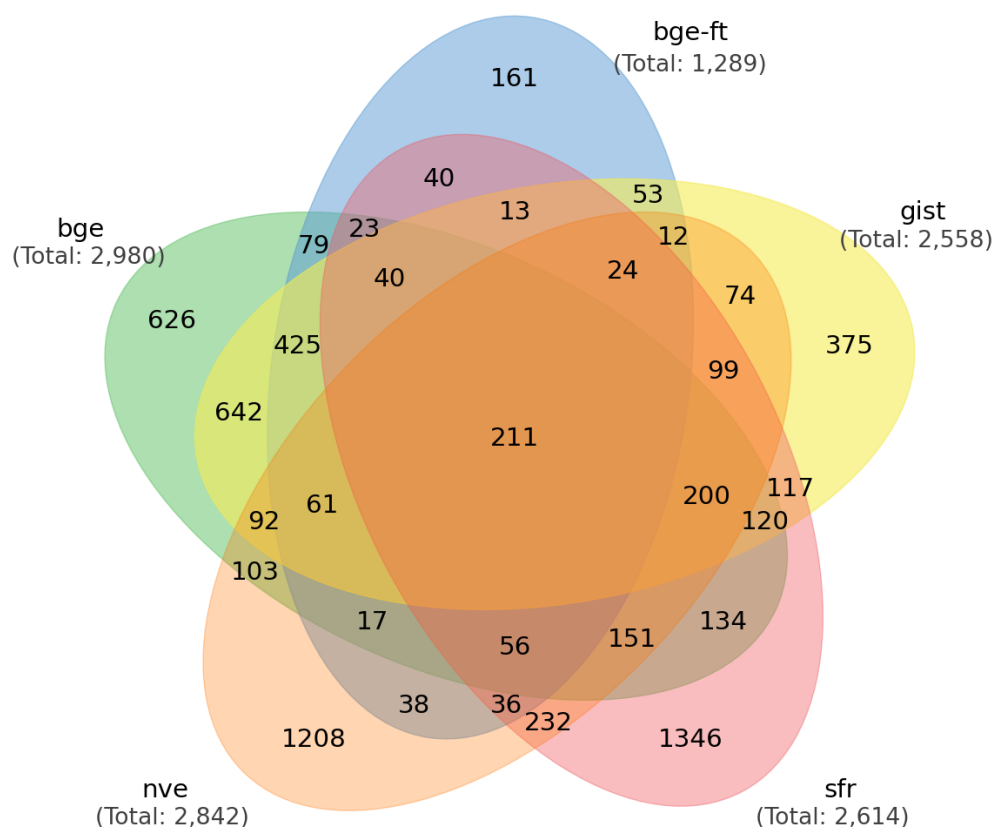


Figure 4.8: Venn diagram illustrating the overlap of misclassified pairs on the test dataset between all five embedding models and the four finalist classifiers. The central intersection shows 211 pairs were misclassified by all models, indicating shared data challenges.

calculus-based physics courses covering modern and classical mechanics, respectively.

Their shared title and subject matter create high semantic overlap, but they are not equivalent for transfer purposes (Appendix C.2). It is worth noting that such an intra-institutional comparison is an unlikely scenario for a framework designed primarily for inter-institutional articulation.

- **Ambiguous or Vague Course Descriptions:** This occurs when descriptions are too brief or use generic language, lacking the specific detail needed for differentiation. This

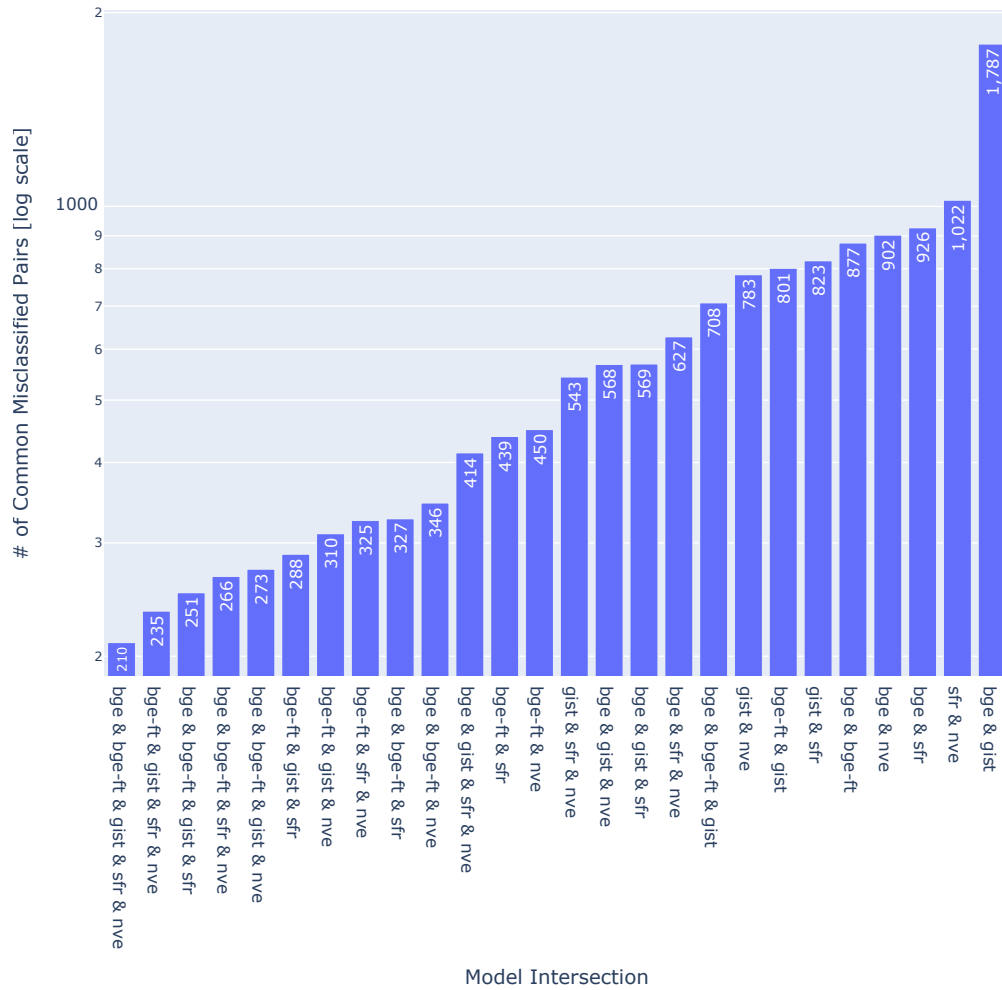


Figure 4.9: Number of common misclassified pairs on the test dataset between all five embedding models and the four finalist classifiers. The high counts of common errors across different models point to similar challenges inherent in the data.

is a known challenge in the field of short-text semantic similarity, where a lack of rich context inherently increases ambiguity [4]. For example, the description for COMM-1 at Saddleback College uses abstract language like “processes of communication” and “development of ideas,” which could be semantically close to a wide range of introductory courses in public speaking, rhetoric, or even philosophy (Appendix C.3).

Qualitative Analysis of False Negatives

A False Negative (FN) occurs when the system fails to identify a true equivalence, representing a missed opportunity that could cause a student to unnecessarily retake a course. These errors are predominantly a consequence of inconsistencies and information gaps in the source data itself:

- **Semantic Divergence in Descriptions:** This error occurs when two officially equivalent courses are described using vastly different terminology, phrasing, or pedagogical focus. For example, a pair of courses with the same C-ID (CDEV-100) are described differently: one (Bakersfield College’s CHDV-B21) uses the language of traditional developmental psychology (“psychosocial”, “milestones”), while the other (Cerritos College’s CD-110) uses the language of social justice pedagogy (“diversity and inclusion,” “anti-bias curriculum”). The model correctly assesses the texts as semantically dissimilar; the failure lies in the inconsistency of the source data, where the ground truth asserts an equivalence not present in the text (Appendix C.4).

- **Incomplete or Minimalist Descriptions:** This error arises when one or both course descriptions in an equivalent pair are too sparse to provide sufficient textual signal for the model to establish a confident match. An example is a pair of Physics courses (both C-ID PHYS-205), where one description from Moorpark College is rich with keywords like “kinematics” and “Newtonian mechanics,” while the corresponding description from Imperial Valley College is a brief, high-level survey summary, lacking the conceptual depth for the model to map the two courses closely (Appendix C.5).
- **Data Quality and Labeling Errors:** In some cases, a false negative can be traced back to a fundamental error in the ground-truth data. A striking example is a pair of courses labeled with C-ID SOCI-125, where the description for one course is for an LGBTQ+ studies class, while the description for the other is for an introduction to statistics. This is almost certainly a data entry error in the source corpus. The model correctly identifies the courses as non-equivalent but is penalized with a false negative because the ground-truth label is erroneous (Appendix C.6).

4.7 Summary

This chapter presented a comprehensive, multi-stage empirical evaluation that validated the proposed decoupled, deep metric learning framework. The investigation began by establishing a performance baseline using a direct Large Language Model approach, which, while capable, highlighted considerable limitations that motivated the development of the

primary pipeline. The subsequent validation of the framework’s core components proved highly successful. Our ablation study confirmed that the novel composite distance vector, Δ_c , is a demonstrably superior feature representation.

The most consequential performance gain was achieved through domain-specific fine-tuning. The resulting BGE-ft model, adapted to the PPM Corpus using a `BatchSemiHardTripletLoss` objective, was shown to be statistically superior to all off-the-shelf models, providing strong empirical evidence for the efficacy of applying deep metric learning to engineer a bespoke embedding space for the course catalog domain. In the final evaluation stage, while all four finalist classifiers achieved exceptionally high F_1 -scores, a nuanced trade-off between peak accuracy and computational efficiency emerged. Statistical analysis identified the Support Vector Machine (SVM) as the most accurate and consistent classifier, whereas Random Forest and XGBoost proved to be significantly more efficient, making them compelling alternatives for real-world deployment.

Finally, a qualitative misclassification analysis provided a essential perspective on the system’s performance. The analysis revealed that despite the high accuracy of the optimized pipeline, the remaining errors are not random but are largely systemic, stemming from challenges inherent to the data itself, such as semantic divergence in the descriptions of equivalent courses, ambiguity from vague descriptions, and fundamental data quality errors. The comprehensive evaluation, therefore, concludes that while the proposed framework is highly effective, the primary bottleneck for further improvement likely does not lie in the model architecture or classification algorithm, but in the quality and consistency of the

source data.

Chapter 5

Discussion, Future Work, and Conclusion

This chapter transitions from the empirical validation of the proposed framework to a broader discussion of its implications. Having systematically evaluated each component and demonstrated the model's high performance in Chapter 4, this chapter now seeks to interpret the significance of these results in the context of the course articulation problem. The objective is to discuss the key findings, acknowledge the inherent limitations of the study, and outline promising avenues for future research that build upon this work. The chapter will culminate in a final conclusion, summarizing the primary contributions of the thesis and reiterating its significance for fostering a more equitable and efficient higher education ecosystem.

5.1 Discussion of Results

The empirical results presented in Chapter 4 offer a robust validation of the core hypothesis of this thesis: that a decoupled, deep metric learning framework can overcome the limitations of prior automated approaches to course articulation. This section discusses the significance of these findings, focusing on the vindication of the proposed framework, the critical impact of domain-specific fine-tuning, and the resulting shift in focus from model-centric to data-centric challenges.

An Effective and Robust Framework

The proposed pipeline successfully addresses the distinct challenges that have hindered previous attempts at automation. By relying exclusively on publicly available course catalog data, the framework is a privacy-conscious alternative to enrollment-based methods like *course2vec*, which are constrained by their need for sensitive, proprietary student records and are not generalizable to institutions with no prior transfer history [47, 61]. Furthermore, by decoupling semantic representation from classification, the framework is substantially more scalable, efficient, and interpretable than using large language models for direct, end-to-end classification. It avoids the high computational costs, “black box” opacity, and prompt sensitivity associated with direct LLM approaches while still providing a quantifiable similarity score for each pair. Finally, its use of deep contextual embeddings represents a fundamental advance over older statistical methods like TF-IDF, which lack any true semantic under-

standing and cannot grasp synonymous or related concepts [3].

The Impact of Domain-Specific Fine-Tuning

Perhaps the most prominent finding from the experimental evaluation is the statistical superiority of the fine-tuned **bge-ft** model over all off-the-shelf competitors, including those that are orders of magnitude larger. This result provides powerful evidence that for specialized domains, targeted adaptation is more effective than sheer scale. General-purpose models, despite being trained on vast swaths of the internet, lack the specialized “vocabulary” to appreciate the fine-grained distinctions in course catalog text. For example, they may not understand the subtle but critical differences between a “survey” course, an “introductory” course, and a “foundations” course. The process of fine-tuning with a triplet loss objective effectively retrained the model’s attention mechanism, teaching it the specific semantics and nuances of the academic domain. This allowed it to generate a far more discriminative embedding space, ultimately leading to higher accuracy in the downstream classification task.

The Bottleneck Has Shifted from Model-Centric to Data-Centric

With the optimized pipeline achieving F_1 -scores exceeding 0.98, the framework has pushed the limits of what can be achieved with the available data. The qualitative misclassification analysis in Section 4.6 revealed that the vast majority of the remaining errors

are not due to failures in the model’s semantic understanding. Instead, they are artifacts of the source data itself. The model fails when officially equivalent courses are described with vastly different pedagogical language (semantic divergence) or when descriptions are too vague or minimalist to contain a clear signal. In these cases, the model is performing correctly—it accurately reports that the texts are not semantically similar. The error lies in the ground-truth expectation that an equivalence should be found where none is textually supported. This leads to a critical insight: the primary bottleneck for achieving near-perfect automation has shifted from being model-centric to data-centric. Simply using a larger or more complex model is unlikely to resolve these data-inherent issues.

This finding serves as a convincing, domain-specific case study for the Data-Centric AI (DCAI) paradigm [79, 67]. Championed by distinguished figures like Andrew Ng, DCAI posits that for many practical applications, once a sufficiently robust model architecture is established, the most consequential and efficient path to performance improvement comes from systematically engineering the data itself [18, 62]. This involves improving label consistency, cleaning noisy examples, and ensuring the dataset accurately reflects the problem domain [79, 76]. The results of this thesis validate the DCAI philosophy: state-of-the-art performance was achieved with a well-chosen but not exotic model, and the remaining errors are artifacts of the source data. Further model-tweaking is unlikely to resolve these data-inherent issues; the most promising path forward is to improve the data itself. This suggests that the most promising path to further improvement lies not in novel architectures, but in methodologies that directly address the quality and consistency of the input data [23], a

point that will be explored further in the following sections.

5.2 Limitations of the Current Study

While the proposed framework represents a pronounced advance, it is essential to acknowledge the limitations that define the boundaries of the current study. These limitations, primarily rooted in the nature of the data and the scope of the task, provide critical context for the results and inform the directions for future work.

Data Quality as a Performance Ceiling

The primary limitation, as identified in the discussion, is that the framework’s performance is fundamentally capped by the quality and content of the public course descriptions. The system can only analyze the text that is provided; it cannot infer information that is absent. As the misclassification analysis demonstrated, when course descriptions are vague, minimalist, or use semantically divergent language to describe functionally equivalent courses, the model’s ability to determine a correct match is severely hindered. This reliance on the source text means the system is vulnerable to inconsistencies and information gaps in how institutions write and publish their catalogs.

Generalizability of the Fine-Tuned Model

The **bge-ft** model was fine-tuned and evaluated on a corpus drawn exclusively from California’s public colleges and universities. While the framework itself is general, the specific fine-tuned model has been specialized for the linguistic patterns, terminology, and pedagogical styles common to this system. Its performance may not be as high “out-of-the-box” on data from private institutions or different state systems or non-domestic systems, which may have distinct catalog writing conventions. Achieving similar performance in a new institutional context would likely require re-tuning the model on a sample of local data.

Scope of Equivalency

This research operationalizes course equivalency as a function of the semantic similarity of their catalog descriptions. This is a powerful proxy, but it does not encompass the full spectrum of factors that human articulation officers may consider. Decisions made by faculty and administrators can be influenced by factors beyond the written content, such as the rigor of the assessment methods, specific lab equipment, or overarching institutional agreements. The current model is not designed to capture this external, non-textual context.

Handling of Complex Articulation Rules

The framework simplifies the articulation task into a binary classification of course pairs (equivalent or not-equivalent). This approach does not natively handle the more complex

articulation scenarios that exist in practice, such as one-to-many (e.g., one university course is equivalent to two community college courses), many-to-many, or non-symmetrical agreements. While the similarity scores produced by the system could inform the discovery of such relationships, the current classification pipeline is not designed to identify them directly, a challenge that persists for many automated systems [46].

5.3 Future Work

The findings and limitations of this study give rise to several promising avenues for future research. These directions aim to address the remaining challenges by improving the quality of the input data, expanding the framework’s capabilities, and exploring more advanced modeling techniques.

First of all, the framework itself can be expanded to be more useful and tunable for institutional needs. In fact, development is currently underway to evolve the framework beyond binary classification and build a full-scale course recommendation engine, which is being actively researched and developed by our team and will provide a conversational interface for a more intuitive user experience. Such a system would use the vector similarity scores to provide students and advisors with a ranked list of the most likely equivalent courses at a target institution. Furthermore, future work could investigate the classifier’s decision threshold as a mechanism for institutional control, allowing administrators to tune the system’s behavior from a more lenient stance to one that errs on the side of caution. To

address the limitation of complex articulations, another avenue of research involves modeling curricula as graphs and applying graph neural networks to identify one-to-many and many-to-many relationships.

In addition, there remain opportunities to refine the core machine learning components of the pipeline. Future work could systematically investigate alternative composite distance measures and feature combinations to determine if a more optimal representation exists for the downstream classifiers. This could be complemented by exploring multi-modal learning, extending the model to analyze not just the catalog description but also the full text of course syllabi or textbook lists. Likewise, more room still remains to explore task-optimized loss functions for fine-tuning. Because the embedding vectors are not being used directly, but as upstream feature engines, creating loss functions that behave more similarly to the downstream classifier may have a significant impact on training efficacy. As models evolve, more sophisticated training methods, such as instruction-tuning on a small, expert-curated dataset, could also be employed to teach the model the explicit task of explaining course equivalency, potentially yielding more interpretable results.

Given that data quality was identified as the primary performance bottleneck, a promising future direction involves implementing a Human-in-the-Loop (HITL) Machine Learning system [72]. Rather than simply having an expert review ambiguous cases, a formal HITL system would create an iterative process that promotes continuous improvement [31, 69]. In this paradigm, the model would flag low-confidence predictions and route them to a human articulation officer. The expert's decision would not only resolve the immediate case but

would also generate a new, high-quality, expert-verified labeled data point. This new data can then be fed back into the training set to periodically re-fine-tune the model, making it progressively smarter and more aligned with expert judgment over time [58, 68]. This approach directly addresses the data-centric bottleneck while building institutional trust by positioning the AI as a tool that augments, rather than replaces, human expertise.

To address the limitation of complex articulation rules, a promising direction for future research is to model the problem using Graph Neural Networks (GNNs) [33, 25]. An academic curriculum is not a mere collection of courses; it is an intricate graph where courses are nodes and relationships like prerequisites are the edges that connect them [80, 71]. Standard NLP models, including the one in this thesis, treat course descriptions as isolated documents, ignoring this rich structural information. GNNs, by contrast, are explicitly designed to learn from both the features of the nodes (e.g., the text embedding of a course) and the structure of the graph itself [66, 27]. A GNN-based approach would learn an embedding for each course that is a function of both its own textual content and the embeddings of its neighbors in the curriculum graph. This structurally-aware embedding could capture the understanding that “Calculus I” followed by “Calculus II” at a community college forms a sequence that, together, might be equivalent to a single “Calculus for Engineers” course at a university. This would allow the system to directly identify these complex, multi-course articulation pathways, as a compelling step beyond the current framework’s capabilities [74, 80].

5.4 Conclusion

The manual process of determining course equivalency remains a substantive impediment to student mobility in higher education, creating administrative burdens and systemic inequities that lead to student credit loss and delayed graduation [64, 9]. This thesis confronted this challenge by designing, developing, and validating a novel computational framework that successfully automates course articulation using only publicly available data. The work’s primary contribution is a privacy-preserving, scalable, and computationally efficient pipeline that overcomes the limitations of previous automated approaches. This was achieved through two key technical innovations: the application of deep metric learning to fine-tune a bespoke embedding model for the specific semantics of the academic domain, and the design of a novel composite distance vector that provides a richer feature set for downstream classification.

The result of this research is a highly accurate framework, capable of achieving state-of-the-art performance on a real-world dataset. More importantly, it represents a practical tool that institutions can use to reduce administrative workload, provide faster and more consistent guidance to students, and ultimately foster a more transparent and equitable educational ecosystem. By mitigating the barriers faced by transfer students, particularly those from underrepresented backgrounds [63], this work contributes a meaningful step toward fulfilling the promise of accessible and efficient pathways through higher education.

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Appendix A

LLM Prompts

For all prompts < and > enclose variables.

A.1 Example: Topic Extraction

Below is a template for the entire prompt used for topic extraction. “⟨” and “⟩” enclose variables.

```

1  You are given the task of parsing course information from raw text.
   This text may be in formats that are hard to read or are incorrectly
   formatted. Your task is to correctly separate the course description
   into its component parts. These parts are the course department code,
   course number, title, unit count, lecture hours, key topics covered,
   the appropriate student audience, the prerequisite requirements, and
   available alternate grading options. Finally, with all the information
   you are given, deduce the academic discipline that the \textbf{course b
   } belongs to. Your audience is another expert, so omit terms or words
   that are irrelevant to the course. The following is the text you are
   analyzing:
2  “‘<course description>’”
3
4  The output should be formatted as a JSON instance that conforms to the
   output schema below. As an example, for the schema {‘‘properties”: {‘‘

```

```

foo": {'title': 'Foo', 'description': 'a list of strings', 'type':
  'array', 'items': {'type': 'string'}}}, 'required': ['foo']} the
  object {'foo': ['bar', 'baz']} is a well-formatted instance of the
  schema. The object {'properties': {'foo': ['bar', 'baz']}} is not
  well-formatted. Here is the output schema:
5  '''<output schema generated by Pydantic>'''
6
7  **Consider conjunctions and disjunctions:** Pay attention to
  conjunctions (e.g., 'and') and disjunctions (e.g., 'or') for all
  categories. These indicate that multiple conditions may be required,
  and they should be reflected in the output.
8
9  **Be exhaustive:** Ensure that all possible conditions or options are
  considered and included in the output. Do not exclude any valid options
  or conditions.
10
11 **For the 'student audience category' field:**
12 * **Identify the target phrase:** Look for phrases like 'Primarily
  for', 'Intended for', 'Designed for', etc., that indicate the
  intended student audience.
13 * **Extract relevant terms:** Once the target phrase is found,
  extract the subsequent words that describe the student majors or fields
  of study.
14 * **Exclude unnecessary words:** Filter out any irrelevant words such
  as 'students', 'majors', 'and', etc., that do not provide specific
  information about the student audience.
15
16 **For the 'alternate grading options' field:**
17 * **Identify keywords:** Look for keywords such as 'Grades', '
  Grading', 'Option', etc., that indicate the presence of alternate
  grading options.
18 * **Extract options:** Extract the specific grading options mentioned
  , such as 'P/NP', 'Letter Grade', 'Pass/Fail', etc.
19 * **Exclude unnecessary words:** Similar to the 'students' field,
  remove unnecessary words like 'options', 'available', etc.
20
21 **Do NOT speculate** Ensure that you use ONLY the information provided
  and do not attempt to infer any information if it is not explicitly
  provided EXCEPT for the academic discipline.
22
23 Respond only with valid JSON. Do not write an introduction or summary.

```

A.2 Example: Raw Text Equivalency Evaluation

```

1 Determine whether the two following courses are equivalent. For course
  equivalency, respond with <response_type>.
2
3 Course 1:
4 '''<target>'''
5
6 Course 2:
7 '''<source>'''
8
9 The output should be formatted as a JSON instance that conforms to the
  output schema below.
10
11 As an example, for the schema \{"properties": \{"foo": \{"title": "Foo
  ", "description": "a list of strings", "type": "array", "items": \{"
  type": "string"\}\}\}, "required": \["foo"\]\} the object \{"foo": \["
  bar", "baz"\]\} is a well-formatted instance of the schema. The object
  \{"properties": \{"foo": \["bar", "baz"\]\}\} is not well-formatted.
12
13 Here is the output schema:
14 '''
15 <output schema generated by Pydantic>
16 '''
17
18 *Consider conjunctions and disjunctions:**
19 Pay attention to conjunctions (e.g., "and") and disjunctions (e.g., "or
  ") for all categories. These indicate that multiple conditions may be
  required, and they should be reflected in the output.
20
21 **Be exhaustive:** Ensure that all possible conditions or options are
  considered. Do not exclude any valid options or conditions.

```

A.3 Example: Topic Equivalency Evaluation

```

1 User: As an expert course description evaluator you are given the task
  of determining whether the two following courses are equivalent.
2 The course data you are given is separated into categories:
3     1. Discipline
4     2. Course Title
5     3. Topics (formatted as a list)
6 The "Target Course" describes the candidate course that the source
  course may fulfill the requirements for.
7 The "Source Course" describes the course that a student has taken that
  may fulfill the requirements for the target course.
8

```

```

9   For course equivalency, respond with 0 if it does not fulfill the
    requirements, 1 if it does fulfill the requirements, 2 if you are
    unsure, or 3 if there is insufficient data.
10
11  Target Course:
12  '''
13  1. <target["discipline"]>
14  2. <target["title"]>
15  3. <target["topics"]>
16  '''
17
18  Source Course:
19  '''
20  1. <source["discipline"]>
21  2. <source["title"]>
22  3. <source["topics"]>
23  '''
24
25  The output should be formatted as a JSON instance that conforms to the
    output schema below.
26
27  As an example, for the schema \{"properties": \{"foo": \{"title": "Foo
    ", "description": "a list of strings", "type": "array", "items": \{"
    type": "string"}}}, "required": ["foo"]\} the object \{"foo": ["bar", "
    baz"]\} is a well-formatted instance of the schema. The object \{"
    properties": \{"foo": ["bar", "baz"]\}\} is not well-formatted.
28
29  Here is the output schema:
30  '''
31  <output schema generated by Pydantic>
32  '''
33
34  *Consider conjunctions and disjunctions:**
35  Pay attention to conjunctions (e.g., "and") and disjunctions (e.g., "or
    ") for all categories. These indicate that multiple conditions may be
    required, and they should be reflected in the output.
36
37  **Be exhaustive:** Ensure that all possible conditions or options are
    considered. Do not exclude any valid options or conditions.
38
39  **EXAMPLES:**
40  EXAMPLE 1:
41  Target Course:
42      1. Mathematics
43      2. Calculus with Analytic Geometry I
44      3. ["Calculus", "Analytic Geometry", "Differentiation", "Integration
    ", "Limits", "Curve Sketching", "Applications"]
45  Source Course:
46      1. Mathematics

```

```
47     2. Calculus I
48     3. ["Calculus", "Differentiation", "Integration", "Transcendental
functions"]
49 Response: 1
50
51 EXAMPLE 2:
52 Target Course:
53     1. Computer Science
54     2. Programming Concepts and Methodology II
55     3. ['Application of object-oriented techniques', 'Program complexity
using abstraction', 'Algorithm analysis and Big-O notation', 'Advanced
language features', 'Basic sorting and searching algorithms', '
Recursion']
56 Source Course:
57     1. Computer Science
58     2. Introduction to Object-Oriented Programming: Java
59     3. ['Object-oriented design', 'Software engineering principles', '
Fundamental programming skills in Java']
60 Response: 0
```


Appendix B

ROC/AUC Curves and Confusion Matrices

B.1 ROC/AUC Curves

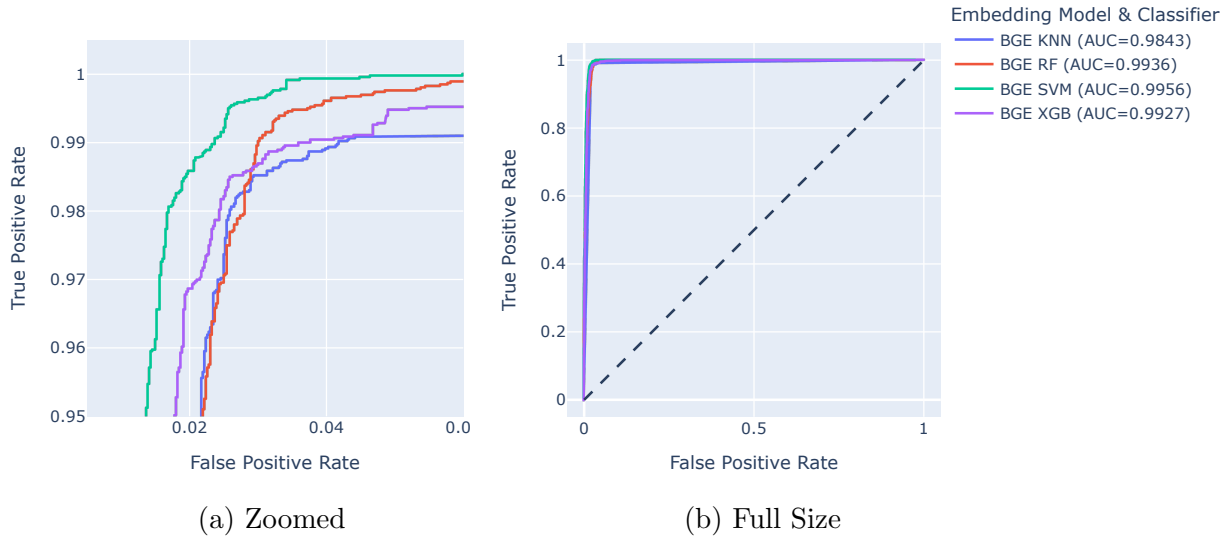


Figure B.1: BGE ROC Curves for Finalist Classifiers with Test Data

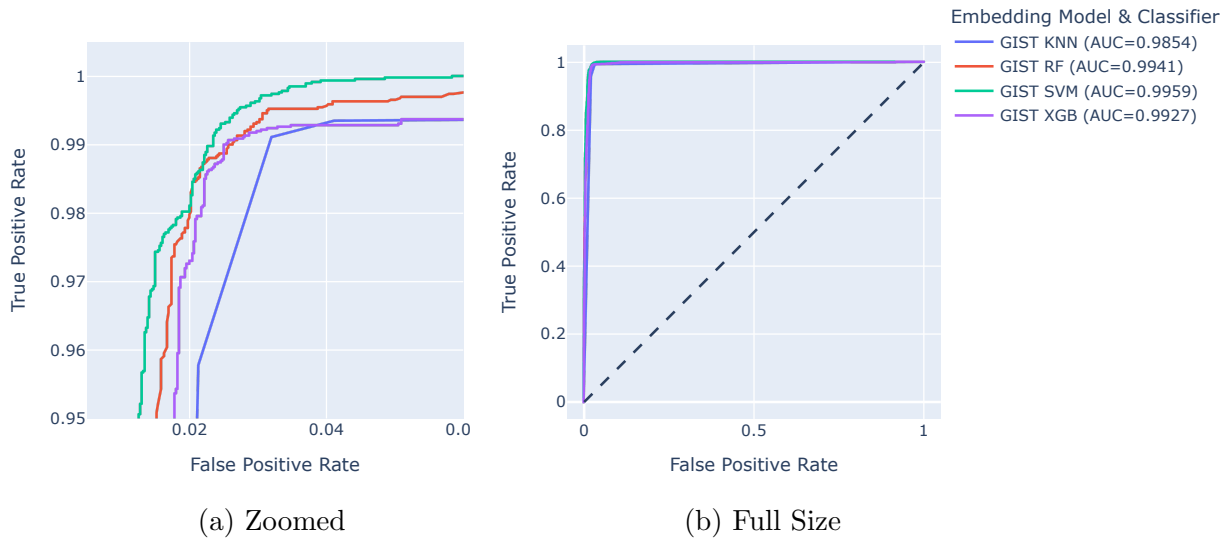


Figure B.2: GIST ROC Curves for Finalist Classifiers with Test Data

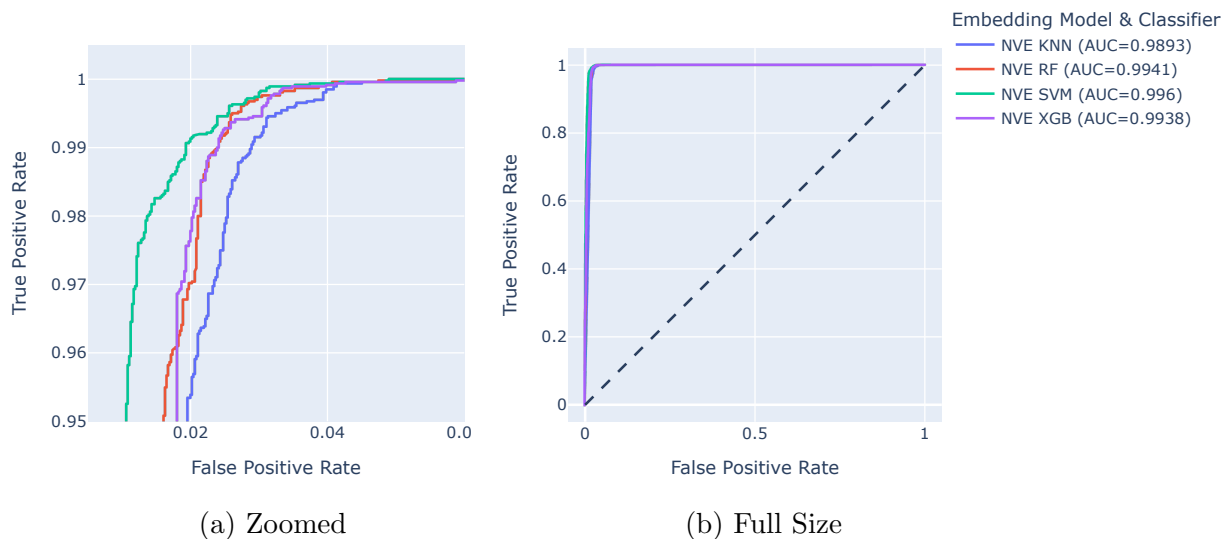


Figure B.3: NVE ROC Curves for Finalist Classifiers with Test Data

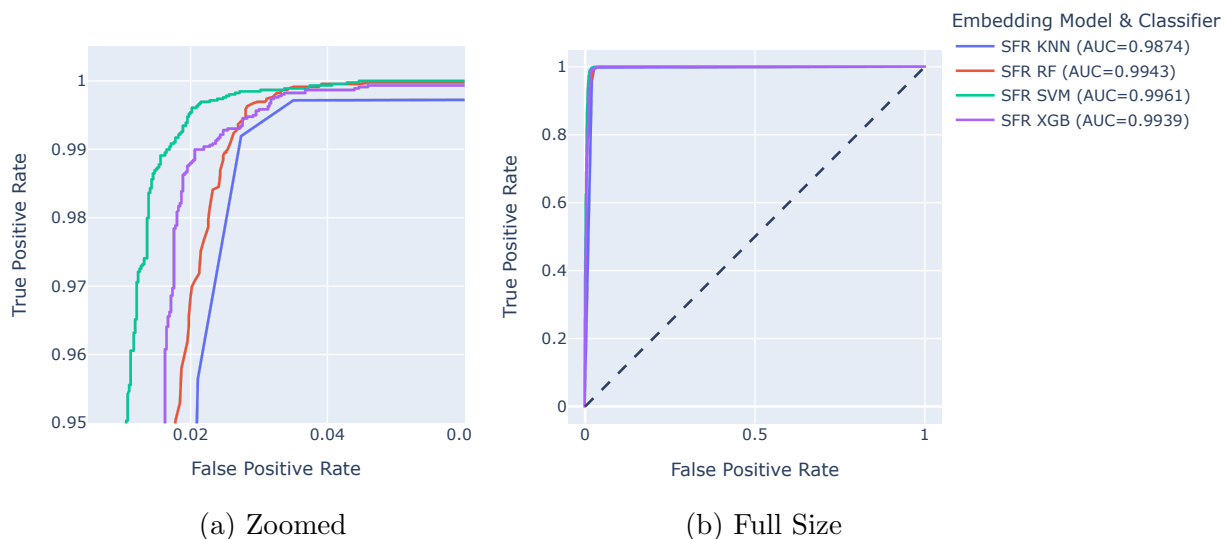


Figure B.4: SFR ROC Curves for Finalist Classifiers with Test Data

B.2 Confusion Matrices

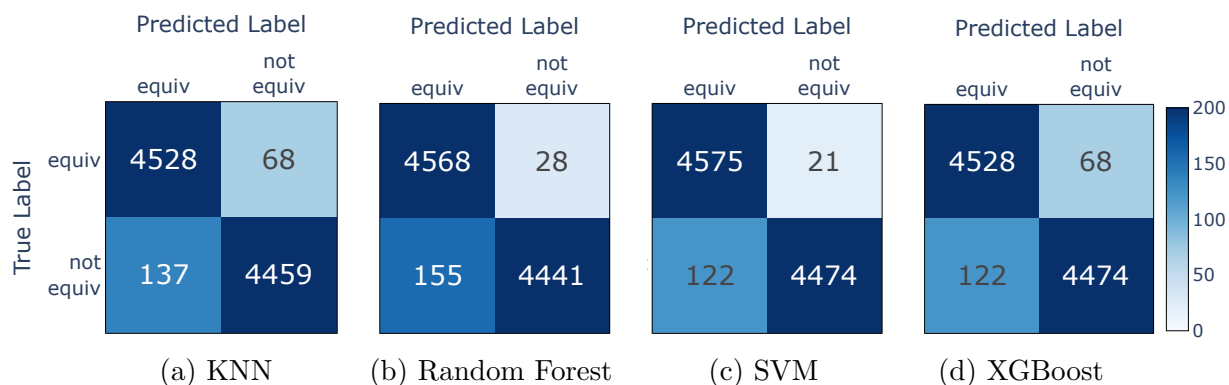


Figure B.5: BGE Confusion Matrices for Finalist Classifiers with Test Data

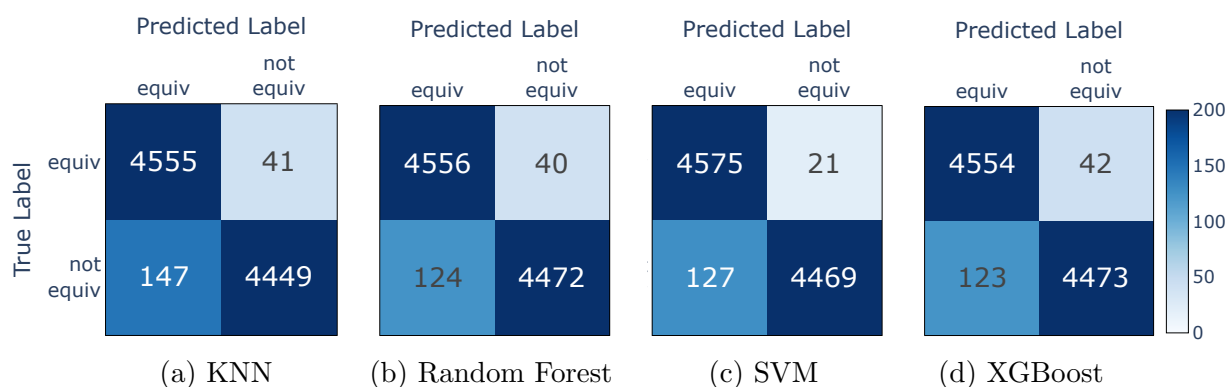


Figure B.6: GIST Confusion Matrices for Finalist Classifiers with Test Data

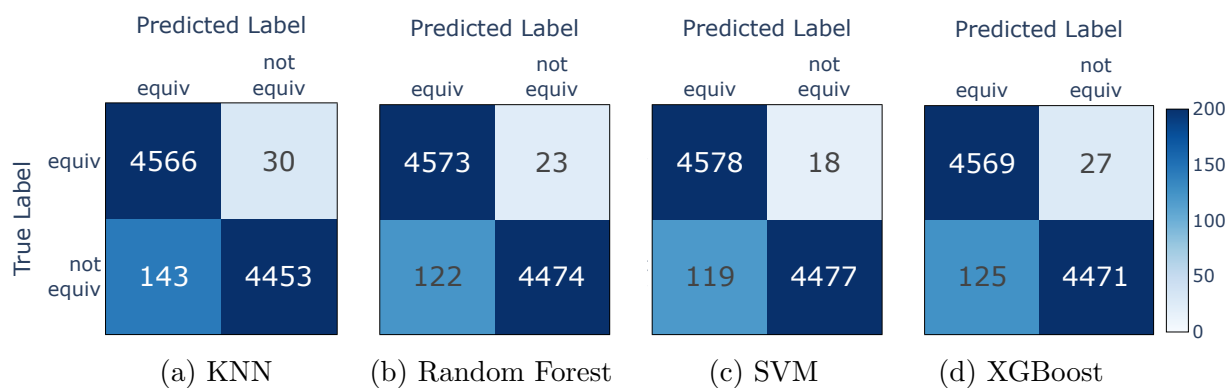


Figure B.7: NVE Confusion Matrices for Finalist Classifiers with Test Data

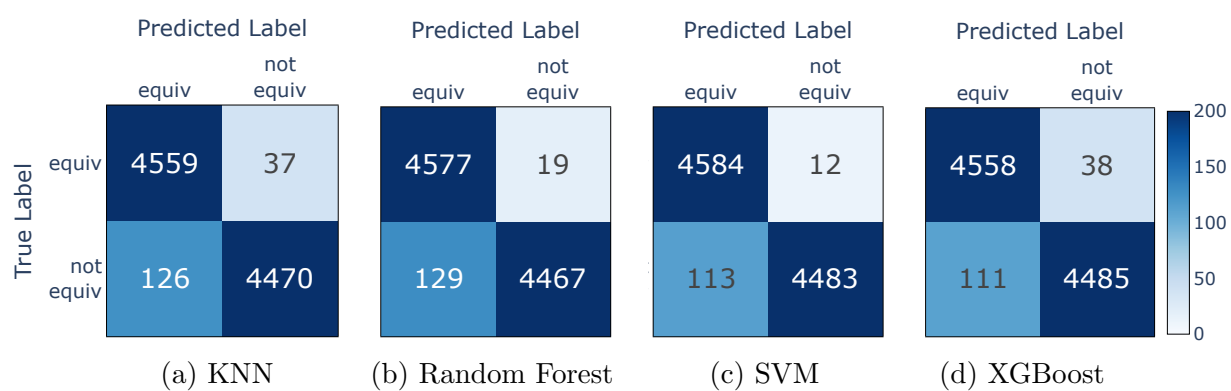


Figure B.8: SFR Confusion Matrices for Finalist Classifiers with Test Data

Appendix C

Misclassification Analysis

This appendix provides the full, unabridged text of representative misclassified course pairs discussed in Section 4.6, allowing for a direct, in-depth review of the error types.

C.1 Most Misclassified Courses

This is an inspection of the top 5 most misclassified courses across all classifiers and embedding models for both the training and test sets.

Table C.1: Most Misclassified Courses (Training)

Rank	Course
1	CID: ENGL-120 Santa Barbara City College ENG-222 Survey of British Literature Offered only in the Spring semester. Survey of British literature during 1798-present, including fiction, poetry, drama and essays. Prerequisite: ENG 110 or ENG 110H

Table C.1: Most Misclassified Courses (Training)

Rank	Course
2	CID: JOUR-130 Fullerton College JOUR-271 F Introduction to Spanish-Language Reporting Advisory: Understanding of conversational Spanish. 54 hours lecture and 18 hours lab per term. This course will guide students in the methods and styles of reporting and writing in Spanish for print and online. It will prepare students to publish stories and photos on the campus' Spanish-language publication. The course also provides students with a general understanding of contemporary Spanish-speaking and Latino communities. (CSU) (Degree Credit)
3	CID: PHYS-140 Berkeley City College CHEM-30A Introductory General Chemistry Fundamental principles of general chemistry: Metric measurements, matter and energy, atomic structure, chemical nomenclature, chemical bonding, chemical reactions, stoichiometry, gas laws, nuclear chemistry; properties of liquids, solids, solutions, acids, and bases.
4	CID: SOCI-125 Compton College SOCI-120 Introduction to Statistics and Data Analysis for the Behavioral Sciences This course introduces students to lesbian, gay, bisexual, transgender, and queer studies. It is designed to analyze power, privilege, and oppression in connection to current macro and micro dynamics in society. Students will evaluate significant historical LGBTQ+ events that fostered changing society's views on sexual identities, as well as, events that have worked against advancing LGBTQ+ rights. In addition, students will evaluate various theories and concepts that attempt to explain the social construction of gender, sex, and sexuality. Furthermore, students can analyze socialization, cultural values, norms, intersectionality, and the unique challenges that impact LGBTQ+ communities.
5	CID: PHYS-200 Foothill College PHYS-4A General Physics (Calculus) Mathematics-physics interrelationships, classical Newtonian mechanics. Lecture and lab hours indicated are per week. This course includes 5 lecture hours and 3 lab hours per week.

Table C.2: Most Misclassified Courses (Test)

Rank	Course
1	CID: CDEV-100 Cerritos College CD-110 Child Development Examines the historical and current perspectives on diversity and inclusion and the impact of systemic societal influences on children's development, learning, and school experiences. Strategies for developmentally, culturally, and linguistically appropriate anti-bias curriculum will be explored as well as approaches to promote inclusive and anti-racist classroom communities. Includes self-reflection on the influence of teachers' own culture and life experiences on teaching and interactions with children and families. Transfer Credit: CSU; UC
2	CID: ECE-230 Cerritos College CD-124 Teaching in a Diverse Society Introduces the appropriate use of assessment and observation tools and strategies to document young children's development and learning. The use of findings to inform and plan learning environments and experiences are emphasized. Recording strategies, rating systems, portfolios, and multiple assessment tools will be discussed, along with strategies for collaboration with families and professionals. Transfer Credit: CSU
3	CID: COMM-110 Saddleback College COMM-1 Communication Fundamentals Understand and use the processes of communication in making personal and social decisions in everyday life, including an understanding of problems and propositions; organization and development of ideas; evidence; methods of research, criticism and evaluation. Presentation of ideas in informative and persuasive contexts. Platform speaking experience will be required (formerly SP 1).
4	CID: CDEV-100 Cypress College PSY-145 C Child Psychology This course explores the physical, cognitive, communicative/linguistic, and socio-emotional development of the child from conception through adolescence across diverse cultures with an emphasis on the learning process. Education and teaching issues related to children are highlighted. (UC/CSU, AA GE, CSU GE, IGETC, CDEV 100)

Table C.2: Most Misclassified Courses (Test)

Rank	Course
5	CID: MUS-180 East Los Angeles College MUSIC-501 College Choir This course is an introduction to choral ensemble singing. Emphasis is on vocal technique and choral elements such as blend, intonation, diction, and music reading. Repertoire is chosen on the basis of group ability and represents historical and current styles of music. Students are required to perform in a public performance at the end of the semester.

C.2 False Positives: Topical Overlap

This error occurs when courses cover the same broad subject but differ in critical dimensions such as academic level or their position in a curricular sequence. The model correctly identifies high semantic similarity but fails to capture the nuanced distinctions (e.g., Part I vs. Part II) that make the courses non-equivalent for articulation.

Table C.3: FP Examples: Topical Overlap

Course A	Course B
Evergreen Valley College PHYS-004B (C-ID: PHYS-210) General Physics This course is one of a three-semester series in calculus-based general physics, serving students majoring in engineering, chemistry, physics, mathematics and other sciences. It emphasizes conceptual aspects of electricity, magnetism, circuits, and Maxwell's equations, and requires quantitative analysis of real world situations.	Foothill College PHYS-4C (C-ID: PHYS-200) General Physics (Calculus) Thermodynamics; mechanical, acoustical, and electromagnetic waves; optics. Lecture and lab hours indicated are per week. This course includes 5 lecture hours and 3 lab hours per week.

Table C.3: FP Examples: Topical Overlap

Course A	Course B
East Los Angeles College ENGLISH-205 (C-ID: ENGL-160) English Literature I	East Los Angeles College ENGLISH-206 (C-ID: ENGL-165) English Literature II
In this course, students read, discuss, and analyze major works of English literature from the Anglo-Saxon period to the late eighteenth century, to develop an understanding and appreciation of the poetry, fiction, and drama of these literary periods and to express that appreciation in a critical analysis.	This course surveys British Literature from the late 18th century emergence of the Romantics, such as Blake, Wordsworth, Coleridge, Byron, Shelley, and Keats; through the Victorian Era, writers such as Browning, Tennyson, Austen, Stevenson, Wilde, and Shaw; and into the early twentieth century, the rise of Modernism and after, writers such as Conrad, Eliot, Yeats, Woolf, Joyce, and Beckett.
Ventura College PSY-V01 (C-ID: PSY-110) Introduction to Psychology	Cabrillo College PSYCH-4 (C-ID: PSY-150) Introduction to Biological Psychology
This course provides an overview of the scientific study of psychology in the areas of neuroscience, sensation and perception, states of consciousness, learning and memory, intellect and cognition, language, lifespan development and the influences of heredity and environment on behavior, motivation, sexuality, emotion, personality, stress and coping, psychological disorders, psychotherapy, and social relations.	Introduces the scientific study of the biological bases of behavior and its fundamental role in the neurosciences. Physiological, hormonal, and neurochemical mechanisms, and brain-behavior relationships underlying the psychological phenomena of sensation, perception, regulatory processes, emotion, learning, memory, and psychological disorders will be addressed. The course also notes historical scientific contributions and current research principles for studying brain-behavior relationships and mental processes.

C.3 False Positives: Ambiguous or Vague

This error occurs when course descriptions are too brief or use generic, high-level language. This lack of specific detail increases ambiguity and can cause the model to identify semantic similarity between courses that are functionally distinct.

Table C.4: FP Examples: Ambiguous or Vague

Course A	Course B
Cabrillo College BUS-18 (C-ID: BUS-120) Business Law	Long Beach City College LAW-18 (C-ID: BUS-125) Fundamental of Business Law
Introduces the United States justice system, covering and relating criminal, civil, employment, torts and contract laws to business operations. History and nature of law, court systems, administrative agencies, crimes, cyber law, the formation and operation of contracts, corporate organization structures, ethical decisions and corporate responsibility and antitrust laws will be covered.	Formerly LAW 18A. This course is designed to explore the overall fundamental understanding of business law today. It examines the scope of how contracts and tort law affect the civil legal process as well as the nature of our current business environment. It is appropriate for students who wish to pursue a career in the business field.
Cerritos College COMM-132 (C-ID: COMM-140) Fundamentals of Small Group Communication	Saddleback College COMM-1 (C-ID: COMM-110) Communication Fundamentals
As an introduction to the fundamentals of group discussion, this course explores small group communication theories to examine group development; leadership in groups; group communication norms, and processes with emphasis on problem-solving, decision-making, and conflict-reduction techniques. Students will learn a variety of techniques to prepare and deliver group presentations.	Understand and use the processes of communication in making personal and social decisions in everyday life, including an understanding of problems and propositions; organization and development of ideas; evidence; methods of research, criticism and evaluation. Presentation of ideas in informative and persuasive contexts. Platform speaking experience will be required (formerly SP 1).

C.4 False Negatives: Semantic Divergence

This error occurs when two officially equivalent courses (i.e., sharing a C-ID) are described using vastly different terminology or pedagogical focus. The model correctly assesses the texts as semantically dissimilar; the failure lies in the inconsistency of the source data, where the ground truth asserts an equivalence that is not clearly supported by the text.

Table C.5: FN Examples: Semantic Divergence

Course A	Course B
Bakersfield College CHDV-B21 (C-ID: CDEV-100) Child Growth and Development: Birth Through Adolescence	Cerritos College CD-110 (C-ID: CDEV-100) Child Development
This introductory course examines the major physical, psychosocial, and cognitive/language developmental milestones for children, both typical and atypical, from conception through adolescence. There will be an emphasis on interactions between maturational processes and environmental factors. While studying developmental theory and investigative research methodologies, students will observe children, evaluate individual differences and analyze characteristics of development at various stages.	Examines the historical and current perspectives on diversity and inclusion and the impact of systemic societal influences on children's development, learning, and school experiences. Strategies for developmentally, culturally, and linguistically appropriate anti-bias curriculum will be explored as well as approaches to promote inclusive and anti-racist classroom communities. Includes self-reflection on the influence of teachers' own culture and life experiences on teaching and interactions with children and families.
Cerritos College CD-124 (C-ID: ECE-230) Teaching in a Diverse Society	Oxnard College ECE-R107 (C-ID: ECE-230) Teaching in a Diverse Society

Table C.5: FN Examples: Semantic Divergence

Course A	Course B
Introduces the appropriate use of assessment and observation tools and strategies to document young children’s development and learning. The use of findings to inform and plan learning environments and experiences are emphasized. Recording strategies, rating systems, portfolios, and multiple assessment tools will be discussed, along with strategies for collaboration with families and professionals.	This course examines the impact of various societal influences on the development of children’s social identity. Students encounter that diversity is a major cultural trait of the United States, and recognize that schools reflect the societal makeup of our country. The course includes an identification of the main differences and similarities among various cultural groups and those of the mainstream culture.
Santa Barbara City College SOC-106 (C-ID: SOCI-160) Sociology of Deviance	Fullerton College SOC-292 F (C-ID: SOCI-160) Introduction to Criminology
Examination of deviance and social control in contemporary society, using the sociological perspective. Focuses on the social processes involved in the construction of deviance, and its functions and impacts on individuals and society. Covers interpersonal and family violence; mental disorders; deviant sexuality; drug and alcohol use; and property, white-collar and organized crime.	This course is a study of theories of crimes and criminal behavior, including an explanation of crime, its causes, and how crime is measured. Major sociological and social science theories will be explored surrounding the issues of crime and criminal behavior.

C.5 False Negatives: Incomplete or Vague

This error arises when one or both course descriptions in an equivalent pair are too sparse or generic to provide sufficient textual signal for the model to establish a confident match. Unlike semantic divergence where both descriptions are detailed but different, here the issue is a lack of substantive, comparable information.

Table C.6: FN Examples: Incomplete or Vague

Course A	Course B
Cypress College GEOG-202 C (C-ID: GEOG-160) Field Geography - Physical	Palomar College GEOG-195 (C-ID: GEOG-160) Regional Field Studies in Geography
Each separate offering of this course will occur in a unique location, studying unique circumstances and conditions in the field. Each course will employ its own combination of technical equipment, scientific instruments, and geotechniques. All courses will study the basic conceptual materials, with modifications associated with the location and the specific conditions encountered at each season.	Extended field studies of the geography of selected regions. Emphasis upon field observation and interpretation of climate, meteorology, vegetation, soils, and landforms.
Imperial Valley College PHYS-200 (C-ID: PHYS-205) General Physics I	Moorpark College PHYS-M20A (C-ID: PHYS-205) Mechanics Solids and Fluids
This course is designed to give an understanding of the fundamental principles of physics in the area of mechanics.	Introduces the basic principles of the mechanics of solids and fluids. Uses calculus to develop the subject matter. Covers kinematics, Newtonian mechanics including rotational dynamics, work, energy, fluid statics and dynamics, and simple harmonic motion.

C.6 False Negatives: Data or Labeling Errors

This error arises when there is a data entry error or the label is incorrect.

Table C.7: FN Examples: Data Entry or Labeling Error

Course A	Course B
Santa Barbara City College PSY-150 (C-ID: SOCI-125) Statistics for the Behavioral Sciences	Compton College SOCI-120 (C-ID: SOCI-125) Introduction to Statistics and Data Analysis for the Behavioral Sciences
Principles and procedures of measurement, data analysis, probability, sampling theory and statistical significance. Covers basic descriptive statistics, including measures of linear relationships and standard scores. Covers the logic of hypothesis testing and inferential statistics up to analysis of variance, including a conceptual introduction to factorial designs. Students conduct analyses by hand and computer software.	This course introduces students to lesbian, gay, bisexual, transgender, and queer studies. It is designed to analyze power, privilege, and oppression in connection to current macro and micro dynamics in society. Students will evaluate significant historical LGBTQ+ events that fostered changing society's views on sexual identities, as well as, events that have worked against advancing LGBTQ+ rights. In addition, students will evaluate various theories and concepts that attempt to explain the social construction of gender, sex, and sexuality. Furthermore, students can analyze socialization, cultural values, norms, intersectionality, and the unique challenges that impact LGBTQ+ communities

Appendix D

Code Base and Documentation

D.1 Fine-tuning Algorithm

Algorithm 1: Simplified Fine-Tuning Process

```

1: procedure FINETUNE_MODEL(base_model_name, train_path, eval_path, config)
2:                                     ▷ 1. Initialization
3:   train_data ← LOAD_DATA(train_path)
4:   eval_data ← LOAD_DATA(eval_path)
5:
6:   model ← GET_MODEL(base_model_name)
7:   loss_function ← INITIALIZE_LOSS(config.loss_type)
8:
9:                                     ▷ Data Handling
10:  sampler ← INITIALIZE_SAMPLER(GroupByLabelBatchSampler, train_data)
11:  train_dataloader ← INITIALIZE_DATALOADER(train_data, sampler)
12:  evaluator ← INITIALIZE_EVALUATOR(BinaryClassificationEvaluator, eval_data)
13:
14:                                     ▷ Optimizer and Scheduler
15:  optimizer ← INITIALIZE_OPTIMIZER(AdamW, model.parameters, config.learning_rate)
16:  scheduler ← INITIALIZE_SCHEDULER(config.scheduler_type, optimizer)
17:
18:  best_score ←  $-\infty$ 

```



```

19:
20:                                     ▷ 2. Training Loop
21:   for epoch ∈ 1 . . . config.NUM_EPOCHS do
22:     for batch ∈ train_dataloader do
23:                                     ▷ Forward Pass
24:       embeddings ← model.forward(batch.inputs)
25:
26:                                     ▷ Loss Calculation
27:       loss ← loss_function.calculate(embeddings, batch.labels)
28:
29:                                     ▷ Backpropagation
30:       optimizer.zero_grad()
31:       loss.backward()
32:       optimizer.step()
33:     end for
34:
35:                                     ▷ 3. Evaluation at Epoch End
36:     current_score ← evaluator.evaluate(model)
37:
38:                                     ▷ Checkpoint Best Model
39:     if current_score > best_score then
40:       best_score ← current_score
41:       SAVE_CHECKPOINT(model)
42:     end if
43:
44:                                     ▷ Update Learning Rate
45:     scheduler.step()
46:   end for
47:
48:                                     ▷ 4. Post-Training
49:   final_model ← LOAD_BEST_CHECKPOINT()
50:   return final_model
51: end procedure

```